

Mortgage Refinancing and the Marginal Propensity to Consume

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Abstract

This paper investigates the role of mortgage refinancing in shaping the estimates of marginal propensity to consume (MPC) and its implications for fiscal policy. Using U.S. household data, we find that MPCs decrease during the year of mortgage refinancing and stabilize afterwards, particularly among households with lower liquid assets, higher debt-to-income ratios, and valuable illiquid assets. The empirical evidence suggests that refinancing provides extra liquidity, reducing MPCs. We leverage on a partial equilibrium model to quantitatively assess these effects and to explore the role of home-equity extractions for fiscal policy. Our findings highlight a new dimension for the efficacy of cash transfers: targeted programs that consider higher MPCs of no-refinancers generate savings between 4 and 12% compared to non-targeted programs. These estimates imply approximately \$30 billions in potential savings under the CARES Act of March 2020.

JEL Codes: E21, E62, G21, G51, H31

Keywords: Mortgage Refinance, Households Heterogeneity, Marginal Propensity to Consume, Fiscal Transfers, Housing, Fiscal Policy.

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1 Introduction

A growing literature argues that the composition of households' balance sheets is important for the transmission of macroeconomic policies. In the context of heterogeneous agent models, the heterogeneity in assets holdings of different degree of liquidity generates large and dispersed marginal propensity to consume (MPC) across households (Kaplan et al. (2018), Auclert et al. (2024)), which in turn determines the efficacy of monetary and fiscal policy. In the United States, mortgage debts and housing equities accounts for approximately 80% of homeowners's wealth, while two-thirds of U.S. households own a home (2001 Survey of Consumer Finance). The illiquid nature of home equity can potentially constrain the ability of households in the use of their wealth to smooth shocks of different nature and to maintain stable consumption patterns. However, households have the option to convert a portion of their home equity into liquidity through home-equity extraction - *mortgage refinance* -, which allows consumption smoothing in the case of a temporary drop in income or unforeseen financial needs. To this extent, Hurst and Stafford (2004) find that around two-thirds of every dollar of equity extracted was converted into current consumption during 1991-94, among liquidity constrained households, while Chen et al. (2020) provide evidence of a counter-cyclical component of home equity extraction, in line with a higher demand for liquidity during bad times.

Our goal in this paper is to ask: how does mortgage refinance affect the marginal propensity to consume? What are the implications for fiscal policy, for instance cash transfers, of aggregate cash-out activity? The dynamics effects of mortgage refinance in shaping the distribution of MPCs have received less emphasis from the literature and these changes could potentially impact, in turn, the efficiency of fiscal policy. It is crucial to note that a significant portion of fiscal transfers occurs during recessions or economic downturns, periods when the prevalence of refinancing tends to be higher. This context underscores the particular significance of the questions we are pursuing.

We make two key empirical contributions. First, we leverage on U.S. household-level data from the Panel Study of Income Dynamics (PSID) and apply the method developed by Blundell et al. (2008) to estimate MPCs conditional on the timing of refinance. We unveil notable reductions in the MPC during the year of mortgage refinancing, followed by subsequent stabilization. Additionally, our analysis reveals distinct features among refinancing households, characterized by lower proportions of liquid assets relative to total wealth, higher debt-to-income ratios, and ownership of higher-value illiquid assets.

These observations suggest a predilection among refinancing households for maintaining liquid resources to facilitate consumption smoothing, echoing parallels with [Kaplan et al. \(2014\)](#) depiction of “wealthy” hand-to-mouth households (W-HtM). Such households constitute a substantial share of U.S. middle class, and are often subject to fiscal policy related debate. While previous research posited that such liquidity-constrained households could display large MPC out of transitory income, a recent study by [Chetty et al. \(2020\)](#)¹ argues that the recent fiscal stimulus might have been ineffective, for many households only spent a small fraction of the stimulus checks. Our empirical findings perhaps suggest an alternative narrative: the wealthy hand-to-mouth households indeed face liquidity shortfall, however, they have access to extra liquidity via home equity extraction, an option that does not exist for most poor hand-to-mouth households. An imminent liquidity injection relaxes the budget constraint for the refinancers, hence lowering their MPCs. Our main findings are robust across different specifications conditioning on the distribution of liquid wealth, liquidity needs for large expenses (medical and educational), expected labor income growth, and hand-to-mouth category of households.

In our second empirical contribution we examine the extent of time-varying micro-MPCs induced by mortgage refinancing, by constructing a measure of consumption-to-income elasticity that varies over time with *aggregate* refinancing activity, and we show that it is highly counter-cyclical. Our result suggests that the efficiency of stabilization policies such as discretionary fiscal policy (e.g. cash transfers) can be improved by targeting households who did not refinance their mortgage in the previous periods.

Motivated by these findings, we turn to a partial equilibrium life-cycle model based on [Boar et al. \(2022\)](#), which features uninsurable idiosyncratic income risk, housing, and mortgage choices, with the objective to quantitatively determine the role of mortgage refinancing for fiscal policy. Our quantitative analysis reveals that MPCs derived from cash transfers exhibit significant variations contingent upon the timing of mortgage refinancing, in line with the empirical evidence from the PSID. As an additional dimension of inquiry, our analysis takes into account the unique circumstances of the Covid-19 pandemic, characterized by a surge in mortgage refinancing activities alongside the implementation of stimulus fiscal transfers by the U.S. government (Figure A.3). To this extent, we perform several ad-hoc scenarios with income and interest rate drops and evaluate the efficacy of fiscal transfers: aggregate MPCs are close to 20%, in line with the estimate from [Parker et al. \(2022\)](#) based on the Consumer Expenditure Survey (CEX), while refinancers

¹Opportunity Insight economic tracker data.

have lower MPCs, around 15%. To further gauge the potency of the refinancing mechanism, we show that an alternative version of the model, one that deliberately excludes the option to refinance, generates a lower *aggregate* MPC of 16%, highlighting the importance of refinancing activity in models that feature housing and mortgages, especially in the context of fiscal policy evaluation. Lastly, we show that targeting no-refinancers can generate savings that range between 4 to 12% of untargeted fiscal programs. Our estimates suggest approximately \$30 billions in potential savings with fiscal targeting from the economic stimulus payments of the CARES Act, in the aftermath of the Covid-19 pandemic in March 2020.

The remainder of the paper is organized as follows. Section 2 provides information on the data, empirical methodology and presents the main results. Section 3 presents evidence of counter-cyclical mortgage refinancing, providing a link to fiscal policy. Section 4 describes the theoretical model and discusses the results from the quantitative exercises. Section 5 concludes.

Related Literature Our paper contributes to three strands of literature. First, several works have focused on the role of monetary policy in shaping incentives for households to refinance and smooth consumption. For example, [Wong \(2021\)](#) finds that younger households that refinance or enter new loans display large consumption response to monetary policy shocks. Along similar lines, [Berger et al. \(2021\)](#) and [Eichenbaum et al. \(2022\)](#) show that the impact of monetary policy is state-dependent: on the previous interest rate path for the former, and on the distribution of savings from refinancing for the latter. [Di Maggio et al. \(2017\)](#) find that reductions in mortgage payments driven by resets of adjustable-rates increases households' consumption, with poor homeowners having higher MPCs. Our paper contributes to these works by providing evidence of changes in MPCs due to refinancing activity, which determines the response of consumption to monetary policy and to fiscal policy, which is the focus of this paper.

Second, our paper relates to the extensive recent literature on heterogeneous agents and the role of incomplete markets in determining the distribution of MPCs, mainly attributed to the category of hand-to-mouth households that display low liquid assets holdings and thus high consumption sensitivity to income changes. Several recent studies have highlighted the transmission mechanism of monetary policy ([Kaplan et al. \(2018\)](#)), fiscal policy ([Kaplan and Violante \(2014\)](#), [Auclert et al. \(2024\)](#)) and estimates of intertemporal MPC ([Fagereng et al. \(2021\)](#), [Auclert et al. \(2024\)](#)). Furthermore, [Kaplan and Violante \(2022\)](#) demonstrate the importance of assuming two types of assets (liquid and illiquid) in order

to generate realistic level of aggregate wealth. Strictly related to this frontier, [Boar et al. \(2022\)](#) quantify the degree of liquidity constraints in the U.S. using a life-cycle model with home-equity extraction via refinance. The authors find that four-fifths of U.S. homeowners are liquidity constrained, a share which is significantly higher than hand-to-mouth households. Furthermore, [Chen et al. \(2020\)](#) show that households demand liquidity via home equity extractions during downturns. They estimate a structural model that quantitatively accounts for the evolution of households' balance sheets and consumption. Our paper sheds light into the mechanism through which "wealthy" hand-to-mouth households gain access to extra liquidity by utilizing home-equity extractions when confronted with income shortfalls.

Lastly, our paper is closely related to the extensive body of literature on how changes in house prices influence aggregate consumption ([Mian and Sufi \(2009, 2011\)](#); [Mian et al. \(2013\)](#)). An upswing in house prices motivates households to pursue more frequent refinancing, given the elevated valuation of their homes, which serves as collateral for borrowing. [Berger et al. \(2018\)](#) show that an incomplete markets model with income and house price uncertainty is able to reconcile the empirical evidence of large consumption responses to house price movements. They approximate the consumption response via a "rule-of-thumb" formula with MPC out of temporary income times the value of housing. However, their analysis does not tackle endogenous changes in households' MPCs driven by refinancing activity. Our empirical results show that the micro MPC out of transitory income (the first element of the rule-of-thumb) is conditional on having refinanced the mortgage.

Our paper contributes to these works by accounting for the dynamic effects of mortgage refinancing in shaping the consumption response of liquidity constrained households in the short-run. Lastly, we show that fiscal transfers might not be a one stop solution for all liquidity constrained households.

2 Empirical Analysis

Recent papers have argued that the strenght of the refinancing channel of monetary policy is state-dependent and that it follows the interest cycle that spans over several years (e.g. [Berger et al. \(2021\)](#), [Eichenbaum et al. \(2022\)](#)). To this extent, we contribute to the literature by providing evidence on changes in MPCs driven by refinancing choices, regardless of the position within the interest rate cycle in which they occur. The results are

particularly important in the context of monetary-fiscal policy interaction, since interest rate incentives drive refinancing choices (both in terms of interest costs reduction and access to credit, through payment-to-income constraints) which in turn affect the efficacy of fiscal policy through MPCs.

2.1 Data

We employ PSID data to analyze the impact and short-run dynamics of mortgage refinancing on the marginal propensity to consume of U.S. households. The PSID is a longitudinal survey, started in 1968, that follows households and their offsprings. The survey has been conducted yearly until 1999, when it became bi-annual. The advantage of using PSID data is that the panel dimension allows to control for households-specific effects, while at the same time it provides a large set of variables. In particular, starting from 1999, the survey includes consumption information, together with assets and balance sheets questions.^{2,3}

Our baseline sample covers the years 1999-2021, and we follow closely [Kaplan et al. \(2014\)](#) for data cleaning and sample selection. We use total family income and obtain data on federal and state income taxes directly from the NBER's TAXSIM software in order to calculate disposable income. We provide more information on the construction of consumption, liquid and illiquid assets classifications in Appendix [A.1](#).

We follow the literature on mortgages and housing (e.g. [Chen et al. \(2020\)](#), [Boar et al. \(2022\)](#)) in defining a refinancing event: household at time t has refinanced her mortgage if the percentage change of her total mortgage balance is higher than 5% from period $t - 1$ to period t . At the same time, the household must be a home-owner, have an active mortgage in period $t - 1$ and must exhibit no change in residence. The rationale for the latter condition is to avoid households who refinance their mortgage and change location within the U.S. driven by new professional opportunities. The criteria enable us to isolate alterations in mortgage balances driven solely by the intent to access liquidity through the extraction of home equity. We have a total of 4,305 observations of mortgage refinancing pooled across years. Appendix [A.1](#) provides additional information on the refinancing

²The PSID covers now more than 70% of all consumption items of the Consumer Expenditure Surveys (CEX).

³Compared to other widely used households datasets, the PSID comprises all together the information needed for our research questions. To the best of our knowledge, no other publicly available panel data on U.S. households contains information on consumption, income and wealth composition.

shares, mortgage rates and changes in the overall balance.

Table 1 reports several summary statistics for refinancers and homeowners in 2001. Home-equity extractors do not appear as wealthier compared to the rest of homeowners, according to disposable income and home values, which reassures that the choice to refinance is not particularly driven by house valuations or disperse income processes. At the same time, refinancers display high level of indebtedness in terms of average mortgage balance, debt-income ratios and they hold small amounts of liquid assets. Not surprisingly, the share of “wealthy” hand-to-mouth households⁴ is high (around 45% and 50% respectively) for both refinancers and homeowners.

Table 1: Summary Statistics

	Refinancers	Homeowners
Disposable Income (\$)	77,536 (35,124)	71,869 (35,112)
Net Liquid Assets (\$)	18,400 (48,169)	22,409 (55,022)
Mortgage Balance (\$)	109,632 (70,697)	84,744 (79,116)
Home Value (\$)	222,015 (214,816)	198,687 (215,733)
Net Liquid Assets over Income	0.17 (0.51)	0.30 (0.98)
Debt-to-Income Ratio	1.53 (1.06)	1.10 (0.99)
<i>N</i>	302	2039

Notes: Values are means, with standard deviations in parentheses. Reference year is 2001. Data are deflated by the CPI deflator (2006 base year) where applicable.

Against this background, our analysis aims to detect dynamic changes in MPCs for home-equity extractors which might exhibit, in principle: *i*) high MPC out of transitory income due to their limited liquid assets availability, which in turn could push them to

⁴Kaplan et al. (2014) define households to be “wealthy” hand-to-mouth if they have positive balances of illiquid assets, but they have - on average - less than two weeks worth of disposable income in liquid assets.

refinance their mortgage, and *ii*) given the large cash-out from home equity that can be used to smooth consumption or pay down different expenses (e.g. medical, educational), the MPC out of transitory income is expected to be smaller. We test this hypothesis in our empirical exercises.

2.2 Empirical Approach

To examine how mortgage refinance affects MPCs out of transitory income, we follow the seminal work by [Blundell et al. \(2008\)](#) (BPP henceforth), which we sketch briefly its main steps. Let us define a common income process with orthogonal transitory and permanent components:

$$\Delta y_{i,t} = \underbrace{\eta_{i,t}}_{\text{Permanent}} + \underbrace{\Delta \varepsilon_{i,t}}_{\text{Transitory}} \quad (1)$$

The BPP methodology is a two-stage procedure: first, we regress log income and log consumption expenditures on year and cohort dummies, education, race, family structure, employment, geographic variables, and their interactions with year dummies:

$$y_{i,t} = \alpha^y + \beta_y' \mathbf{X}_{i,t} + \varepsilon_{i,t}^y \quad (2)$$

$$c_{i,t} = \alpha^c + \beta_c' \mathbf{X}_{i,t} + \varepsilon_{i,t}^c \quad (3)$$

The procedure aims at capturing the permanent component of both consumption and income, while the residuals are considered a proxy for the transitory component. We then construct first-differenced residuals of both log consumption $\Delta \hat{\varepsilon}_{i,t}^c$ and log income $\Delta \hat{\varepsilon}_{i,t}^y$. With this specification, BPP estimate the MPC as the coefficient of transitory income shocks to consumption, implemented by IV regression of $\Delta \hat{\varepsilon}_{i,t}^y$ on $\Delta \hat{\varepsilon}_{i,t}^c$, instrumented by $\Delta \hat{\varepsilon}_{i,t+1}^y$:

$$\widehat{\text{MPC}}_t = \frac{\text{Cov}(\Delta \hat{\varepsilon}_{i,t}^c, \Delta \hat{\varepsilon}_{i,t+1}^y)}{\text{Cov}(\Delta \hat{\varepsilon}_{i,t}^y, \Delta \hat{\varepsilon}_{i,t+1}^y)} \quad (4)$$

While the true marginal propensity to consume out of a transitory shock is defined as:

$$\text{MPC}_t = \frac{\text{Cov}(\Delta \hat{\varepsilon}_{i,t}^c, \varepsilon_{i,t})}{\text{Var}(\varepsilon_{i,t})} \quad (5)$$

The estimator $\widehat{\text{MPC}}_t$ is a consistent estimator for the true MPC under the assumption

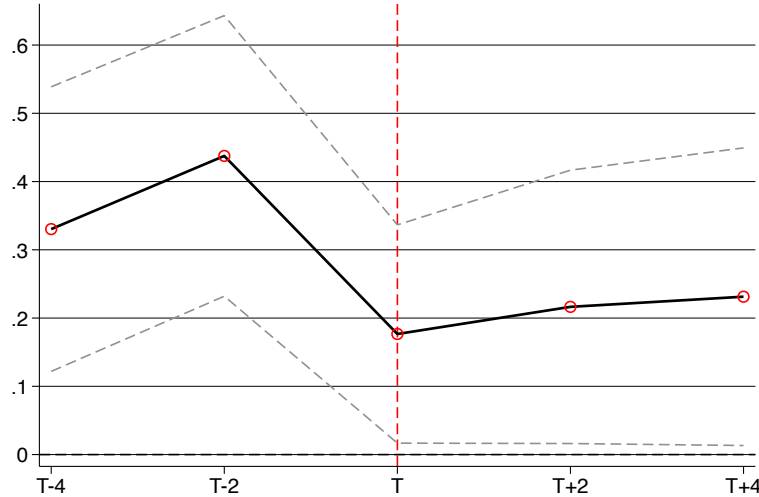
that the household has no foresight, nor advance information about future shocks:

$$\text{cov}(\Delta \hat{\varepsilon}_{i,t}^c, \eta_{i,t+1}) = \text{cov}(\Delta \hat{\varepsilon}_{i,t}^c, \varepsilon_{i,t+1}) = 0 \quad (6)$$

Given our interest in estimating dynamic MPCs, we augment the estimation procedure with a symmetric indicator on the time of refinancing ($\text{Ref}_{i,t}$), from 4 years before refinancing up to 4 years after refinancing. Figure 1 plots the baseline estimates of MPCs.

$$\widehat{\text{MPC}}_t(\text{Ref}_t) = \frac{\text{Cov}(\Delta \hat{\varepsilon}_{i,t}^c, \Delta \hat{\varepsilon}_{i,t+1}^y)}{\text{Cov}(\Delta \hat{\varepsilon}_{i,t}^y, \Delta \hat{\varepsilon}_{i,t+1}^y)} \cdot \text{Ref}_{i,t} \quad (7)$$

Figure 1: (Bi-annual) MPCs Estimates over Home-Equity Extraction



Notes: time T refers to the year of home-equity extraction. Grey dashed lines represents 95% confidence intervals with bootstrapped standard errors based on 250 replications. H_0 : MPC (Before) = MPC (After), p-value= 0.04.

It is important to recall here that the BPP methodology, applied to bi-annual PSID data, implies a consumption share in the first two years of the transitory income shock. The years preceding the mortgage refinancing are associated with relatively high MPCs. In line with our hypothesis, the estimated MPCs suggest that households that extract equity from their home tend to experience tight borrowing constraints and needs for liquidity. At the time of refinancing, and in the years afterwards, the MPCs stabilize at lower (roughly half) values. Our result sheds light on the dynamic impact of households'

choices that affect their balance sheets which influence their MPCs. In particular our estimates highlight the degree of heterogeneity of MPCs across time, which is masked in cross-section estimates for sub-categories of households.⁵

2.3 Robustness Checks

We perform several robustness checks by considering sub-groups of households, in order to validate our hypothesis of home-equity extractions driven by liquidity needs. In practice, we consider the following groups:

- “Wealthy” hand-to-mouth households, according to [Kaplan et al. \(2014\)](#).
- Households with increase on medical or educational spending in the year preceding the mortgage refinancing.
- Liquidity constrained households, based on the criteria set by [Cooper \(2013\)](#): *i*) households below the median of the liquid assets over income ratio distribution; *ii*) households above the median of the debt-service ratio⁶.

The results - reported in Appendix [A.2](#) - confirm that households facing educational or medical expenses in the years before refinancing experience a large drop in MPCs, as well as those that satisfy the conditions of liquidity constraints driven by low liquid wealth income ratios, but not for debt service. Our results seem to highlight the importance of refinancing to tap unexpected expenses and liquidity constraint concerns more than expected future expenses. In particular, in Table [A.2](#) we report the average change in MPCs before and after households undergo a mortgage refinancing: across all different specifications we consider, we confirm large and significant drop in MPCs in the range of 0.20 - 0.26.

Our sample includes historically low interest rates on mortgages, driven in particular by the zero-lower bound period of monetary policy, which might influence the subset of households who decided to refinance by motives that are not related to liquidity needs but by simply reducing the net present value of their debt obligations. Indeed, interest

⁵It is interesting to notice here that the average MPCs before refinancing (around 0.40) is considerably higher than the MPC estimate for “wealthy” hand-to-mouth (0.26). See later section on the intratemporal dimension of MPCs.

⁶Debt service ratio is the ratio of its debt payments (mortgages, home equity debts, car loans and credit cards) over disposable income.

rates incentives (measured as the differential between the mortgage rate contract and current market rates for new mortgages) are one of the fundamental drivers of mortgage refinancing activity (Berger et al. (2021), Eichenbaum et al. (2022)). We show that our main result of decreasing MPCs upon refinancing remains robust to a specification in which we focus solely on households who increased their mortgage balance but do not report a decrease in the interest rate they pay on it. In particular, we are most likely excluding households that received a more favorable interest rate schedule on their mortgage even in cases in which their choice to refinance was driven by liquidity needs, thus we see this specification as the most restrictive one to control for interest rate incentives. Table A.3 shows that the decline in MPCs after refinancing is equal to -0.28, a larger drop compared to the -0.20 in our baseline. The result further strengthens our argument of home equity extraction reducing MPCs for households in particular need of liquidity, a subset of all refinancers.

Furthermore, we confirm that our key empirical result is robust for several other robustness checks based on unemployment, household head selection, cash-on-hand and total consumption (see Appendix A.2).

Lastly, we perform ad-hoc checks related to the BPP methodology and its main assumptions, which we report in Appendix A.3. A key assumption in estimating structural MPCs is the presence of unanticipated income shocks. This assumption is particularly important in the context of mortgage refinancing, as households may anticipate their refinancing decisions in advance. We address this concern in detail in the next section.

2.4 Validity of Identification

It is crucial to address the validity of the no-foresight assumption in the context of the timing of mortgage refinance. First, the assumption relates to potential adjustment in consumption today in anticipation of any income shock (either permanent or transitory) that might occur in future periods. In the case of mortgage refinance, we are assuming that at $t - 1$ (two years prior the home-equity extraction) the household does not adjust (systematically) her level of consumption. In a separate scenario, the agent could foresee a (potentially negative) income shock ahead of time, thus the expected income shock correlates with the option to refinance the mortgage. We assert that such scenario seems implausible and thus the no-foresight assumption in BPP remains valid.⁷ However, we

⁷The literature on household finance and mortgage contracts argue that most households fail to refinance when it is optimally to do so (e.g. for interest rate benefits) and that they delay such option due to

test empirically the validity of our argument by implementing a staggered event-study research design. We follow [Freyaldenhoven et al. \(2022\)](#) in setting our linear panel model with dynamic policy effects:

$$c_{i,t} = \alpha_i + \gamma_t + q'_{i,t}\psi + \sum_{m=-G}^M \beta_m z_{i,t-m} + C_{i,t} + \varepsilon_{i,t} \quad (8)$$

Where α_i represents a households fixed effect, γ_t a time fixed effect, and $q_{i,t}$ a vector of controls. The scalar $C_{i,t}$ denotes a confound that may be correlated with the policy, while $\varepsilon_{i,t}$ is an unobserved shock that is not correlated with the policy. Our dependent variable $c_{i,t}$ is the log of consumption. Lastly, the dynamic effects of the policy (home-equity extractions) are captured by $\sum_{m=-G}^M \beta_m z_{i,t-m}$. The indicator $z_{i,t-m}$ is equal to 0 for all the periods prior refinancing, and equal to 1 afterwards. To facilitate the interpretation, we follow [Freyaldenhoven et al. \(2022\)](#) by modifying the model in (8) by representing it in terms of changes in the policy $z_{i,t}$ and thus cumulative estimated effects. In particular, it is assumed that a change in the policy (event) at time t leads to a change in the outcome L_G periods before $t - G$ and L_M periods after $t + M$. L_G, L_M are non-negative. Specifically, we estimate the following regression model:

$$c_{i,t} = \alpha_i + \gamma_t + \sum_{k=-G-L_G}^{M+L_M-1} \delta_k \Delta z_{i,t-k} + \delta_{M+L_M} z_{i,t-M-L_M} + \delta_{-G-L_G-1} (1 - z_{i,t+G+L_G}) + q'_{i,t}\psi + C_{i,t} + \varepsilon_{i,t} \quad (9)$$

Intuitively, $z_{i,t-M-L_M}$ is an adoption indicator that equals 1 if household i refinanced at least $M + L_M$ periods before period t . Similarly, $(1 - z_{i,t+G+L_G})$ is an indicator for whether household i will adopt the policy more than $G + L_G$ periods after time t . In this way, the parameters $\{\delta_k\}_{k=-G-L_G-1}^{M+L_M}$ are interpretable as cumulative policy effects for different horizons. More precisely, (9) is linked to the model in (8) as follows:

$$\delta_k = \begin{cases} 0 & \text{for } k < -G \\ \sum_{m=-G}^k \beta_m & \text{for } -G < k \leq M \\ \sum_{m=-G}^M \beta_m & \text{for } k > M \end{cases} \quad (10)$$

non-monetary, behavioral factors. We see these arguments in favor of the BPP assumption, validating our results.

Furthermore, we follow the literature by imposing a normalization of the parameter δ_{-1} to zero, which means that the plotted coefficients of the event-study can be interpreted as estimated effects relative to the period before the policy was adopted. Lastly, in a separate exercise we control for cohorts adoptions in the policy, as suggested in [Sun and Abraham \(2021\)](#): under staggered adoption, the estimator β_m can be represented as a weighted average of policy effects for different event times and cohorts. However, [Sun and Abraham \(2021\)](#) shows that the estimated weights can be nonzero (for event-times different than m) and negative (for different cohorts). We implement such scheme that allow for cohort-specific (i.e. first treated cohort and never treated cohort) profiles of policy effects. We include in $q'_{i,t}$ the lagged dependent variable and households characteristics.⁸

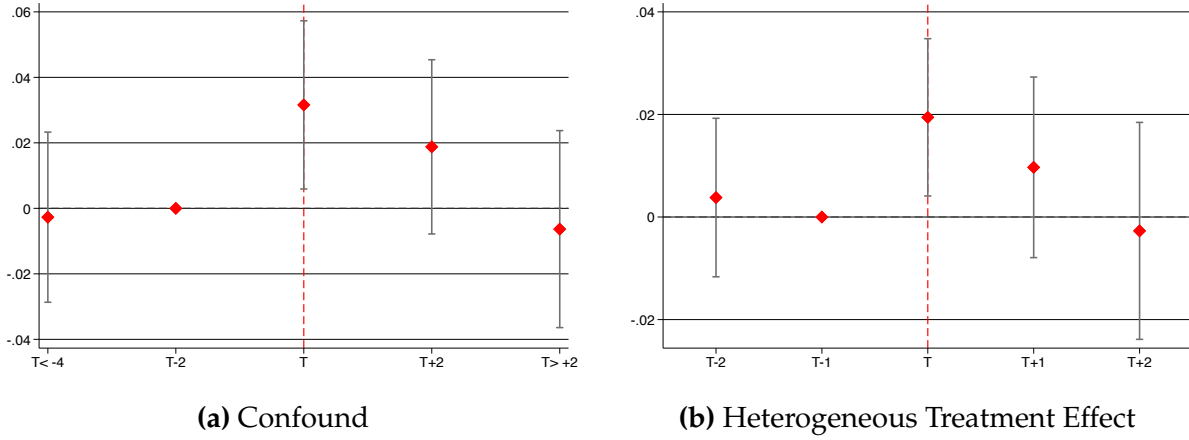
To reduce the endogeneity issue that arises from our setting (liquidity constrained households will be more likely to refinance), we control for confoundedness in the policy by including the ratio of liquid assets over income in $C_{i,t}$: if liquidity needs are associated with home-equity extractions that allows households to smooth (or boost) consumption, we believe that this specification estimates the true dynamic effects on consumption.

Figure 2 plots the estimated δ_k coefficients for **(a)** the homogeneous treatment effect with confoundedness and **(b)** heterogeneous treatment effect as in [Sun and Abraham \(2021\)](#). At the time of refinancing, households increase their non-durable expenditure between 2% and 3%, based on the specification. Most importantly, the point estimates before the home-equity extraction takes place are all not statistically significant, which implies no anticipatory effects and no-changes in consumption that might push households to choose to refinance. Although the coefficient just before the refinancing event is normalize to zero by construction, note that also the coefficient δ_{-2} is not statistically different from zero, which implies no trend or changes in consumption across all the years before refinancing. The results confirm that the decrease in MPCs estimated with the BPP methodology in the years after mortgage refinancing is not associated with anticipated changes in consumption motivated by expected changes in income, which strengthen the validity of our results.

In Appendix A.3.3 we test whether the anticipatory effects on consumption might differ by cohort. Our estimates indicate that only two cohorts (2012 and 2020) adjust positively their level of consumption prior the refinancing event. However, the change in MPCs remains robust to a specification when we exclude these households from our sample.

⁸We include the same control variables and their interactions employed under the BPP estimation.

Figure 2: Dynamic Effects on Log Consumption



Notes: time T refers to the year of home-equity extraction. Red diamonds indicate the point estimates, while grey bars indicate 95% confidence intervals.

2.5 Intratemporal dimension

In the previous section, we underscored the significance of the intertemporal dimension of mortgage refinancing in influencing households' MPCs. We now shift our focus to evaluating the intratemporal aspect of refinancing, which involves estimating the marginal propensity to consume for distinct samples within our dataset, categorized by household type. This approach yields a cross-sectional estimate for each respective category.

Table 2 reveals that households engaging in mortgage refinancing activities exhibit higher MPC when compared to the rest of the sample, particularly in comparison to homeowners and non-refinancers. It is worth noting that the MPC of refinancing households is, however, lower than that of the "wealthy" hand-to-mouth households, which aligns with the rationale that some of the refinancing households face liquidity constraints akin to W-HtM households. Nevertheless, these refinancing households opt for home-equity extraction, thereby mitigating their budgetary constraints.

However, the cross-sectional estimate for refinancers conceals the effects of home-equity extraction in relation to their position within the liquid wealth distribution. To address this, we employ the BPP estimator by conditioning on a sample split for mortgage refinancers, stratifying them based on their liquid wealth relative to their income in comparison to the median value within the sample. Table 3 unveils a substantial divergence in MPCs, notably demonstrating that refinancers positioned at the lower end of

the wealth distribution exhibit a statistically significant higher MPC. This underscores the notion that a significant proportion of refinancers opt for mortgage refinancing primarily due to liquidity requirements. The intertemporal and cross-sectional results are robust for a different measure of liquidity, cash-on-hand, as reported in Appendix A.2.5.

Our findings suggest that cross-sectional estimates of MPCs may obscure the effects of endogenous choices impacting household balance sheets. Specifically, over a four-year period, we demonstrate that mortgage refinancing influences the cross-sectional estimates for homeowners and liquidity-constrained households.

Table 2: (Bi-annual) MPCs over Households Category

	Refinancers	No-Refinancers	W-HtM	Homeowners	Sample
MPC	0.22*** (0.03)	0.17*** (0.03)	0.26*** (0.04)	0.18*** (0.02)	0.19*** (0.02)
N	9,004	11,548	6,687	14,811	20,552

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. W-HtM refers to “wealthy” hand-to-mouth.

Table 3: (Bi-annual) Refinancers MPCs over Liquid Wealth

	Refinancers Liquid Wealth < Median (1)	Refinancers Liquid Wealth > Median (2)
MPC	0.24*** (0.03)	0.16** (0.07)
N	7,059	1,945

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. H_0 : MPC (1) > MPC (2), p-value: 0.86

3 Counter-cyclical Mortgage Refinancing

In this section we show that refinancing activity is particularly relevant in times of downturns and recessions. We first replicate the empirical evidence by [Chen et al. \(2020\)](#) who show that home-equity extraction is counter-cyclical, in line with a higher demand for liquidity. We then construct a novel measure of consumption-to-income elasticity that varies

over time with aggregate refinancing activity in the economy, and we further highlight its counter-cyclical. We contend that these two factors suggest a role for fiscal policy, particularly in terms of targeted cash transfers, which are influenced by heterogeneous MPCs driven by refinancing activities.

3.1 Data

We use data on mortgages originated quarterly in the U.S. via the Fannie Mae Single-Family Loan-Level dataset. We aim to replicate the empirical evidence of [Chen et al. \(2020\)](#), who employ data from the Mortgage Bankers Association, thus we see our exercise as a robust evidence of their results. The dataset tracks each loan monthly from the origination until it is paid off voluntarily or involuntarily via foreclosure. For our purpose, we do not distinguish loans and borrowers, since each loan has a unique identification number, thus we assume that every loan is associated with a new borrower. The sample spans from January 2000 to November 2021 and we keep only 30-year fixed-rate mortgages, which are the most common type of mortgage contract in the United States. Finally, to limit the computational burden, we select a 10 percent random sample of loans originated in each quarter during our sample period.

We define refinanced mortgages those that are prepaid voluntarily before maturity and we focus on total refinancing. We are not able to precisely disentangle mortgage refinancing reasons due to the data structure, but we argue that this does not affect our main implications, as in [Chen et al. \(2020\)](#). However, the Fannie Mae dataset includes information on the purpose of the mortgage at origination: home purchase, interest rate refinance, and cash-out. In Appendix [A.6](#) we confirm that our result is robust when we focus on the share of new cash-out mortgages per quarter.

Since our analysis is on the time dimension of refinancing activity, we construct our refinancing index as the share of prepaid mortgages (N_t^{ref}) over the total number of active mortgage (N_t) in the sample in a given month t : $\text{REFI}_t = N_t^{\text{ref}}/N_t$. We use the variable in logarithm in our analysis. Finally, we include from the FRED database the Industrial Production index IP_t , the Case-Shiller house price index HPI_t , the year-on-year change in the 30-year fixed mortgage rate ΔRM30_t , and as control variables ($\mathbf{X}_t$) we include the 1-year Treasury rate in levels and year-on-year change, and the 30-year fixed mortgage rate. Interest rates enter the specification in percentages, while indices enter in logs.

3.2 Empirical Evidence

We follow [Chen et al. \(2020\)](#) and regress our monthly (log) index of mortgage refinancing on macroeconomic and financial variables as follows:

$$\log(\text{REFI}_t) = \alpha + \beta_1 \log(\text{IP}_t) + \beta_2 \log(\text{HPI}_t) + \beta_3 \Delta \text{RM30}_t + \mathbf{X}'_t \beta_4 + \epsilon_t \quad (11)$$

Table 4 reports the results. Changes in interest rates and house prices are consistent with aggregate refinancing activity: households take advantage of lower interest rates to refinance their mortgages and higher house prices allow them to increase borrowing via higher-valued collaterals. However, the effect of industrial production is negative on refinancing activity, that is households tend to refinance marginally more in downturns, after controlling for interest rate and house price incentives, exactly as in [Chen et al. \(2020\)](#).⁹ We argue that this counter-cyclical evidence of mortgage refinancing generates *aggregate* changes in MPCs, which we proxy through changes in the elasticity of consumption in the next section.

Table 4: Aggregate Refinancing and Economic Activity

Dependent variable: $\log(\text{REFI}_t)$	(1)	(2)	(3)
$\log(\text{IP}_t)$	-3.40*** (0.56)	-3.01*** (0.72)	-2.73*** (0.72)
$\log(\text{HPI}_t)$		0.69*** (0.17)	0.53** (0.25)
ΔRM30_t		-0.46*** (0.05)	-0.46*** (0.07)
Controls	X	X	✓
N	262	262	262
R^2	0.10	0.40	0.48

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.3 Consumption Elasticity

Based on our previous result of counter-cyclical mortgage refinancing, we use the PSID data to estimate and construct a measure of consumption-to-income elasticity that varies

⁹In Appendix A.6 we perform a sample split robustness based upon the Great Financial Crisis, as well as considering the share of cash-out mortgages over the newly originated mortgages in a quarter.

over time with refinancing activity using the following specification:

$$\log c_{i,t} = \alpha + \beta \log y_{i,t} + \delta \text{Ref}_{i,t} + \gamma \log y_{i,t} \cdot \text{Ref}_{i,t} + \varepsilon_{i,t} \quad (12)$$

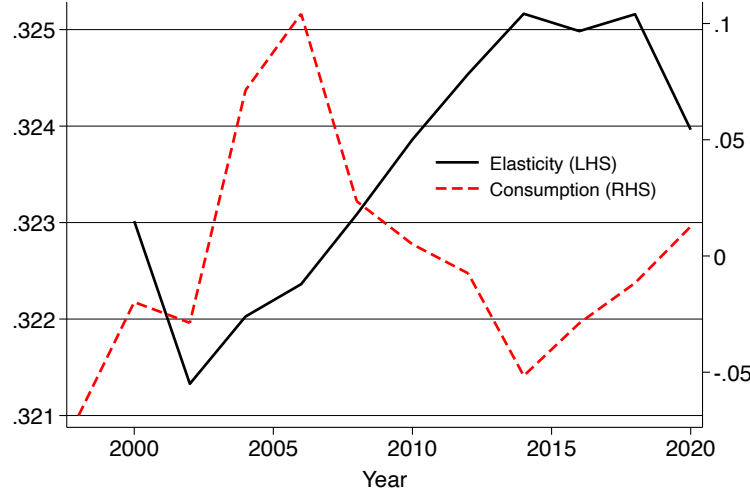
where $\log c_{i,t}$ and $\log y_{i,t}$ are the residualized consumption and income measures following the BPP approach, and $\text{Ref}_{i,t}$ is our previously employed definition of mortgage refinance within the PSID as in Section 2.1. Our consumption-to-income elasticity is defined as:

$$\varepsilon_{i,t}^c \equiv d \log c_{i,t} / d \log y_{i,t} = \beta + \gamma \cdot \text{Ref}_{i,t} \quad (13)$$

The cross-section average of these elasticities varies over time due to changes in the average refinancing activity. Table A.14 in the Appendix reports the regression estimates, which confirms that the sensitivity of consumption to income is reduced when households refinance their mortgage, in line with the structural MPCs estimates of the previous section. The result is robust to additional controls, such as the aggregate refinancing level and interest rate gap (measured as the difference between the average mortgage rates in the PSID and the current market average - nation wide - mortgage rate) in the previous period.

Figure 3 plots the average elasticity together with the detrended aggregate consumption, which indicates a significant counter-cyclical component in $\varepsilon_{i,t}^c$. In periods of high refinancing activity (before the GFC), the elasticity is relatively low, which indicates that consumption is less sensitive to changes in income. The opposite holds for the years following the GFC, characterized by low home-equity extractions and a relatively higher sensitivity until 2020, when refinancing activity spiked in the wake of the Covid-19 pandemic. Although the BPP methodology does not allow to generate MPCs estimates conditioning on the refinancing activity for each single year in the sample, we leverage on the elasticity as a proxy for time-varying MPCs due to refinancing to argue that the efficacy of stabilization tools such as discretionary fiscal policy hinge on the aggregate level of home-equity extractions. We test this hypothesis in a theoretical model with consumption, saving and mortgage choices.

Figure 3: Refinancing Consumption-to-Income Elasticity



Notes: Consumption series (red dashed line) is the cyclical component of average consumption in the PSID estimated via Hodrick–Prescott filter.

4 Model

4.1 Environment

Motivated by the empirical evidence on the relation between MPCs and home equity extraction, we leverage on an overlapping generations model as in [Boar et al. \(2022\)](#) to examine the role of mortgage refinancing for fiscal policy. Individuals in the model have finite lifespans, experience random income fluctuations and housing maintenance shocks, derive utility from consumption and housing, incur disutility from home production, and have the option to save using a short-term liquid asset or by investing in a residential property. The analysis is conducted in partial equilibrium, thus interest rates and housing prices are considered exogenous variables. Individuals face the choice of either renting or owning a home, with the sale of one's property involving a fixed transaction cost. Additionally, individuals can access liquidity by acquiring a mortgage, which incurs both a fixed monetary origination cost and a stochastic utility cost designed to account for non-financial obstacles to borrowing.

Preferences. Agents have life-time utility and live for T periods (each period being a quarter) by choosing consumption c , housing services h , time spent on home production

n , and consume the terminal wealth w .

$$\mathbb{E} \sum_{t=1}^T \beta^{t-1} \left(\frac{c_t^{1-\sigma}}{1-\sigma} + \alpha \frac{h_t^{1-\sigma}}{1-\sigma} - \frac{n_t^{1+\gamma}}{1+\gamma} - \xi_t \right) + \beta^T B \frac{(1+w_{T+1})^{1-\sigma}}{1-\sigma}$$

To obtain a new mortgage (including the option to refinance a pre-existing one), agents incur in a random utility cost ξ . Other elements included in the utility function are the discount factor β , the relative risk aversion σ , the elasticity of home production $1/\gamma$, the weight of housing in preferences α , and the strength of the bequest motive B .

Income. Every period an agent receives income

$$y_{i,t} = \lambda_t z_{i,t} e_{i,t}$$

$$\log(z_{i,t+1}) = \rho_z \log(z_{i,t}) + \sigma_z \varepsilon_{i,t+1}$$

where λ_t is the deterministic life-cycle component of income, $z_{i,t}$ is the persistent component and $e_{i,t}$ is the transitory component of income. The persistent component follows a standard AR(1) process with volatility of innovations σ_z and persistence ρ_z . The transitory component e is iid and drawn from a normal distribution (as the persistent component) with volatility σ_e .

Home production. The supply of hours in home production generate ϕn units of output, which can be used to smooth consumption and aims to capture ways that households provide services to family members.

Liquid assets. Households can save in a standard one-period risk-free asset a with interest rate $r_t(a)$, and borrowing limit $\underline{a} < 0$. The fact that the interest rate is function of the amount of liquid assets holdings aims to capture heterogeneity in rate of return across the wealth distribution (as in [Benhabib et al. \(2019\)](#)).

Housing. Agents who modify the amount of housing stock h to h' priced at p incur in proportional transaction costs $f_s p h$ and $p(h' - h)$ units of output. Additionally, homeowners face maintenance shocks $\delta_t = \bar{\delta}$ with probability π_δ and zero otherwise.

Renting. Agents who do not own a home can rent h units of housing services with rental rate R . By renting, agents are not subject to adjustment costs.

Mortgages. Agents can borrow through mortgages. For each amount borrowed b , the agent needs to make a minimum payment $\bar{m}$, which by no-arbitrage condition is

$$\bar{m} = \frac{r_m}{1 - (1 + r_m)^{-D}} \cdot b$$

where D is the maturity of the mortgage (assumed to be 30 years, in line with most of U.S. mortgages), and r_m is the interest rate on the mortgage. However, agents are free to repay an amount greater than the minimum required ($m \geq \bar{m}$), with no prepayment penalty. Thus the mortgage balance evolves based on

$$b' = (1 + r_m)b - m$$

Mortgage refinancing. The option of refinancing allows homeowners (who do not sell their houses) to extract home equity. The costs associated to close one mortgage and opening a new one are fixed, $f_{0,m}$, and proportional to the new amount borrowed, $f_{1,m}b'$. These costs apply to both cases in which the agent finances the purchase of a new home, or she is refinancing the mortgage on an existing home. Additionally, as specified within the agents' preferences, there are non-monetary costs associated with the opening of a mortgage. These costs (ξ) are iid random draws from an exponential distribution with mean ν and are meant to capture other factors (information, time, behavioral) that reduces the likelihood of refinancing. At the time of opening a mortgage, the agent needs to satisfy a loan-to-value (LTV) constraint on the amount borrowed against the value of the home, and a payment-to-income (PTI) constraint on the minimum mortgage payment over the household's income:

$$\begin{aligned} b' &\leq \theta_m p h \quad (\text{LTV}) \\ \frac{r_m}{1 - (1 + r_m)^{-D}} b' &\leq \theta_y y \quad (\text{PTI}) \end{aligned}$$

Refinancing costs are rolled over into debt, thus agents do not face out-of-pocket costs, and their mortgage balance is equal to

$$\hat{b}' = f_{0,m} + (1 + f_{1,m})b'$$

Budget constraints. In each period, agents in the model choose among five different discrete choices which determine the budget constraint.

An agent who rents will receive the profit from selling her house, net of the selling and maintenance costs, and repays the principal and interests on the outstanding mortgage debt

$$c + a' + R h' = y + \phi n + (1 + r_l(a))a - (1 + r_m)b + (1 - f_s)p h - \delta h$$

An agent who purchase a new home will incur in the housing price and the new mortgage debt (if she chooses to borrow), which includes the fixed and proportional costs, and she is subject to the LTV and PTI constraints.

$$c + a' + ph' - b' = y + \phi n + (1 + r_l(a))a - (1 + r_m)b + (1 - f_s)ph - \delta h$$

Inactive agents simply repay the mortgage payment above or equal the minimum amount determined at origination

$$c + a' = y + \phi n + (1 + r_l(a))a - m - \delta h, m \geq \bar{m}$$

Ultimately, agents who choose to refinance repay the entire original mortgage balance and take a new mortgage subject to the LTV and PTI constraints

$$c + a' - b' = y + \phi n + (1 + r_l(a))a - (1 + r_m)b - \delta h$$

Recursive formulation. In each period the agent will consider and maximize the value function for each option (1. rent, 2. purchase a new home without a mortgage, 3. purchase a new home with a mortgage, 4. refinance an existing mortgage, 5. remain inactive), without considering the utility cost of obtaining a mortgage. By denoting the vector of state variable as $s = (t, a, b, \bar{m}, h, z, e, \delta)$, we can define the envelope over the options by taking into account the iid utility costs:

$$\bar{V}(s) = \int \max(V^1(s), V^2(s), V^3(s) - \xi, V^4(s) - \xi, V^5(s)) dF(\xi)$$

$$F(\xi) = 1 - \exp\left(-\frac{\xi}{\nu}\right)$$

Calibration. We follow the benchmark calibration as in [Boar et al. \(2022\)](#) that replicate salient moments of households portfolio composition, specifically of those who refinance. To this extent, the calibration generates an annual rate of refinancing in the economy of 8%, in line with PSID waves of 1999 and 2001 and with [Bhutta and Keys \(2016\)](#) who use a large panel of consumer credit records. Table 5 reports both external and internally calibrated parameters. In order to account for the dynamic response of MPC for homeowners who decide to refinance, we recalibrate the model - Table 6 - for a separate version that deliberately excludes the option to refinance.

Table 5: Baseline Calibration - Boar et al. (2022)

Assigned	Value	Parameter	Calibrated	Value	Parameter
T	61	Number of years	β	0.985	Discount factor
σ	2	Relative risk aversion	α	0.687	Preference weight on housing
γ	1	Home production elasticity	ϕ	0.939	Efficiency home production
D	30	Mortgage maturity	B	10.26	Bequest motive
r_m	0.025	Mortgage interest rate	R	0.011	Rental rate of housing
θ_m	0.85	Maximum LTV	$f_{0,m}$	1,330	Fixed cost mortgage, 2016 USD
θ_y	0.21	Maximum PTI	τ_0	0.358	Slope liquid int. rate schedule
$f_{1,m}$	0.025	Proportional cost mortgage	τ_1	10.33	Location liquid int. rate schedule
ν	1/3	Mean utility cost mortgage	r_h	0.016	Upper bound liquid int. rate
f_s	0.06	Cost of selling home	ρ_z	0.964	AR(1) persistent income comp.
π_δ	0.10	Prob. maintenance shock	σ_z	0.15	Volatility persistent income comp.
$\bar{\delta}$	0.063	Size maintenance shock	σ_e	0.327	Volatility transitory income comp.
r_l	-0.028	Lower bound liquid int. rate			

Table 6: Alternative Calibration - No Refinance

Calibrated	Value	Parameter
β	0.987	Discount factor
α	0.706	Preference weight on housing
ϕ	0.957	Efficiency home production
B	9.919	Bequest motive
R	0.011	Rental rate of housing
$f_{0,m}$	964.18	Fixed cost mortgage, 2016 USD
τ_0	0.335	Slope liquid int. rate schedule
τ_1	10.93	Location liquid int. rate schedule
r_h	0.0032	Upper bound liquid int. rate

4.2 MPCs of Refinancers

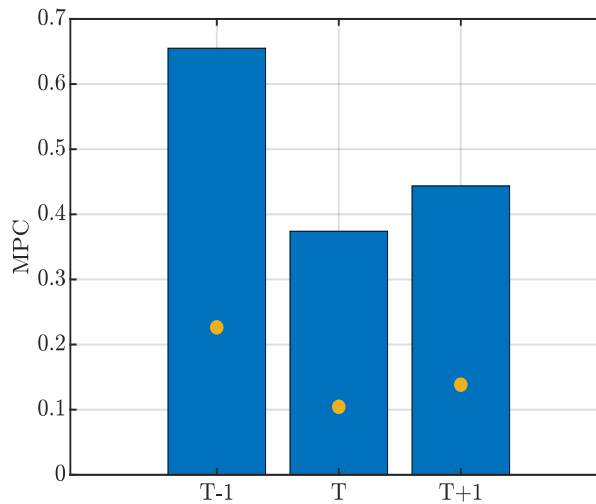
We first simulate the model for several periods, as in the benchmark calibration. We will use this simulated economy as a baseline to assess the impact of all the scenarios we consider in this section. First, we perform an unanticipated one-time \$500 cash transfer to agents that have refinanced at different points in time and calculate their (annual and quarterly) marginal propensity to consume compared to the baseline economy:

$$\text{MPC}_{i,t} = \frac{\tilde{C}_{i,t} - C_{i,t}}{\tau}, \quad i = \underbrace{\{\text{Ref} = t - 4\}}_{T+1}, \underbrace{\{\text{Ref} = t\}}_T, \underbrace{\{\text{Ref} = t + 4\}}_{T-1}$$

where C and $\tilde{C}$ are the consumption levels of households in the economy without and with the fiscal transfer, respectively. The specification enables us to compute the marginal propensity to consume for individuals who have undergone mortgage refinancing one year prior to receiving the fiscal transfer (denoted as $t - 4$). This represents the MPC pertaining to the cash transfer one year after ($T + 1$) the refinancing event. Additionally, we calculate the MPCs for individuals who either refinance at the same time as receiving the transfer or will refinance in a year's time (based on the simulated economy without fiscal transfers). Most importantly, the actual refinance choice does not change with the additional cash transfers (the additional refinancing activity is in the order of 0.2%). Figure 4 illustrates the average MPCs based on the timing of home equity extraction.

Consistent with the empirical findings from the PSID, the annual MPC is notably higher before the refinancing event, and it subsequently decreases and stabilizes after the extraction of home equity. These results underscore two key points: *i*) households opting for refinancing are likely facing liquidity constraints, as they lack sufficient liquid assets to sustain and smoothen their consumption, motivating their decision to refinance; *ii*) the impact of cash transfers on consumption expenditure is contingent on the prevalence of refinancing activities in the economy, given the influence of this activity on micro-level MPCs.

Figure 4: MPCs Conditional on Home-Equity Extraction Timing



Notes: time T refers to the year of mortgage refinance. For instance, $T-1$ refers to the MPC of agents one year before they will refinance. Yellow dots display the first quarter MPCs, and the blue bars the annual MPCs.

To strengthen this argument, we later examine how quarterly MPCs differ in the short

run depending on the timing of the refinancing event. The time dimension enables us to disentangle the cumulative effect presented in Figure 4, facilitating a comparison of the impact of cash transfers over time. These are effectively intertemporal MPCs (iMPCs) as defined by Auclert et al. (2024) for a contemporaneous change in disposable income, which we discuss extensively in Section 4.4.

4.3 Scenarios Analysis

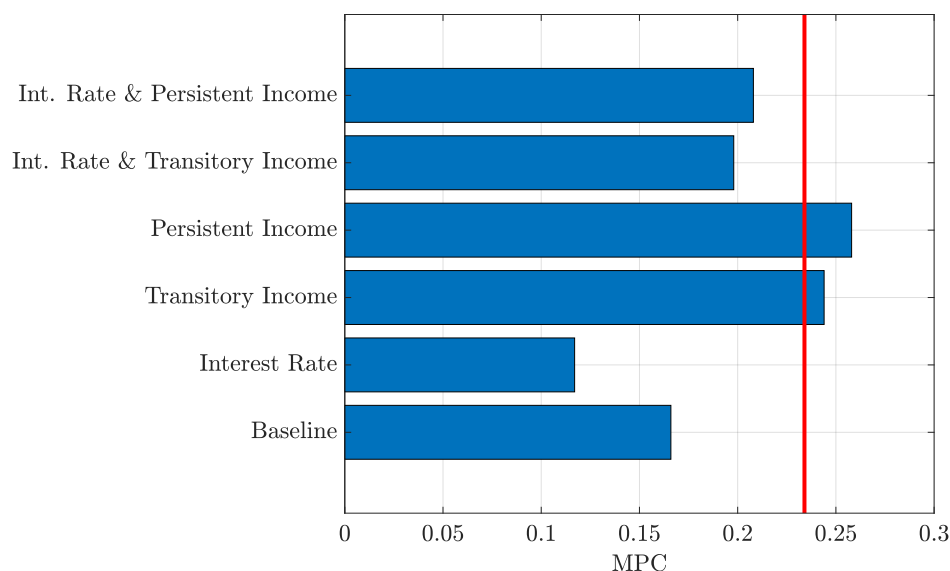
How does the option to refinance the mortgage impact the efficacy of fiscal transfers conditional on downturns? To answer this question, and to relate the model to our empirical results, we simulate separate exogenous scenarios that aim to best approximate the economic conditions of a recession. First, we take as a benchmark the baseline simulation from the previous section to compute the aggregate quarterly MPC. Second, we simulate a permanent drop in the pre-tax interest rate (r_m) of 1.75 percentage points. The reduction in the interest rate alone allows (through the PTI constraints) and incentivizes (through benefits on mortgage costs) more households to refinance. The scenario isolates the implications on MPCs that are solely due to interest rate incentives, along the lines of the recent literature on the effects of monetary policy shocks through the refinancing channel (Berger et al. (2021), Eichenbaum et al. (2022)). Third, we simulate a scenario in which, in a given quarter, all households receive a negative shock realization to their persistent component $z_{i,t}$ as well as - separately - to the transitory component $e_{i,t}$ of income, to replicate the loss in disposable income and unemployment risks. The scenarios aim to capture the liquidity needs faced by households against a loss in their income in order to smooth consumption, with the option to refinance their mortgage. Access to home-equity is, however, limited by the PTI constraint, such that only those with relatively higher income - following the shock - will manage to extract liquidity out of their houses. Lastly, we combine both scenarios which reproduce, as close as possible in the context of the environment of the model, the economic conditions of a recession, for instance as in the Covid-19 pandemic.

Figure 5 shows the aggregate quarterly MPCs under all scenarios. The baseline MPC is on average 0.17, as in the original estimate of Boar et al. (2022). The MPC drops under the reduction of interest rate only (0.12): interest rate incentives reduce (permanently) the total mortgage costs, which in turns allow - ceteris paribus - households to increase their (permanent) consumption. Furthermore, lower interest rates relaxe the PTI constraints for

new home-owners and for agents that opt to refinance. Thus, any additional transitory cash transfer will result less efficient compared to periods in which interest rates were higher.

On the contrary, scenarios of income losses generate higher MPCs than the baseline, 0.26 and 0.24 for persistent and transitory component, respectively. The drop in consumption combined with a lower access to refinancing makes fiscal transfers more efficient: households will spend most of the additional cash in the first quarter, with the average increase in line with the estimates from the literature. Based on the most recent fiscal stimulus in the aftermath of the Covid-19 pandemic, our estimates are very close to [Parker et al. \(2022\)](#) based on CEX data (0.23). Lastly, the joint drop in interest and income generates MPCs that are a linear combination of both independent scenarios: 0.21 and 0.20 for the persistent and transitory income shock that we have considered.

Figure 5: Aggregate MPCs - Simulated Scenarios

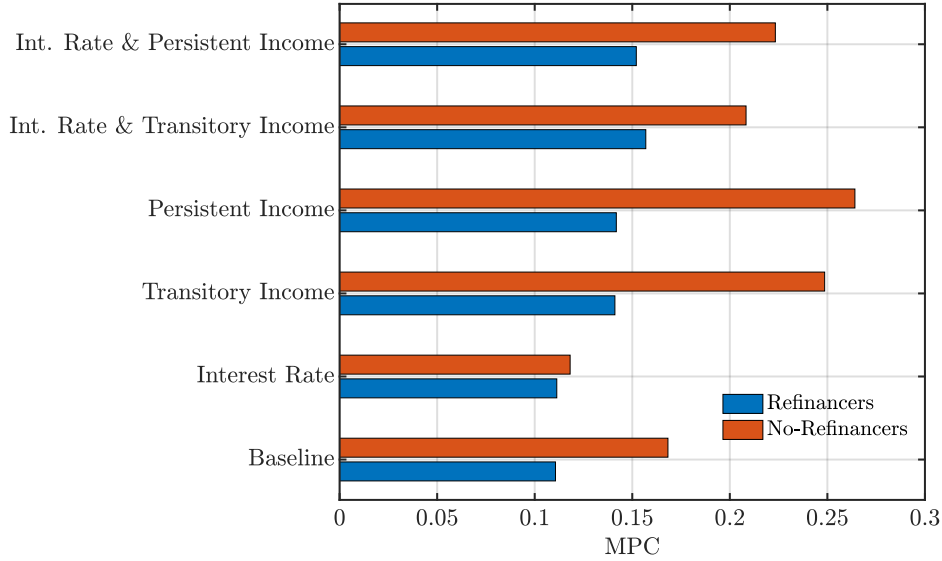


Notes: Quarterly MPCs for each simulated scenarios. The vertical red line represents the average estimate of fiscal transfers from [Parker et al. \(2022\)](#) for all consumption expenditure goods based on the CEX. See the main text for more details.

The results indicate that the nature of the shock matters for the size of MPCs. In particular, our scenarios analysis point towards the importance of aggregate income shocks, which typically arise in a recession, to generate MPCs that are closely in line with the empirical estimates based on micro-data. However, the refinancing choice masks large

heterogeneity in the economy's MPCs: Figure 6 reports the average MPCs of households who refinanced within the preceding year compared to no-refinancing households. The disparity becomes significant in the simulations when income shocks are present, highlighting the larger sensitivity to consumption between the two groups.

Figure 6: Average MPCs - Simulated Scenarios - Refinancers and No-Refinancers



Notes: Quarterly MPCs for each simulated scenarios.

4.4 Intertemporal MPCs

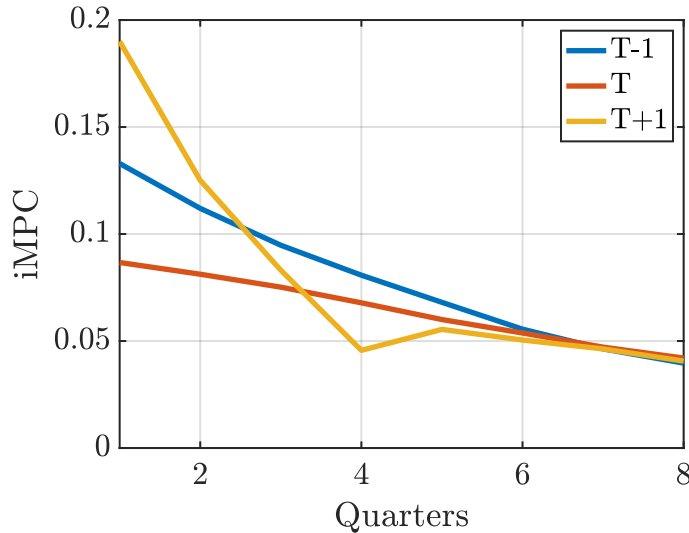
The transmission channel of fiscal policy operates contemporaneously and intertemporally through different propensity to consume out of transfers, as discussed by [Auclert et al. \(2024\)](#). In the context of heterogeneous agents models, the authors show that given the short horizon at which household can plan and smooth consumption, the efficacy of fiscal policy is governed by these intertemporal MPCs (iMPCs), which can be represented in matrix form as the change in consumption at time t conditioning on a transfer occurring at time s :

$$\mathbf{M} \equiv \begin{pmatrix} M_{0,0} & \cdots & M_{0,s} \\ \vdots & \ddots & \vdots \\ M_{t,0} & \cdots & M_{t,s} \end{pmatrix} \equiv \begin{pmatrix} \frac{\partial C_0}{\partial Z_0} & \cdots & \frac{\partial C_0}{\partial Z_s} \\ \vdots & \ddots & \vdots \\ \frac{\partial C_t}{\partial Z_0} & \cdots & \frac{\partial C_t}{\partial Z_s} \end{pmatrix} \quad (14)$$

where $\partial C_t / \partial Z_s$ is the aggregate change in consumption C upon a change in disposable income Z . The iMPCs matrix is a sufficient statistic for the response of macroeconomic variables to fiscal policy, which is useful to estimate general equilibrium path in heterogeneous agents models and fiscal multipliers.

In our analysis we focus on the contemporaneous effect of fiscal transfers - the first column of the M matrix. The contrast in iMPCs between an individual who has recently undergone refinancing and someone who refinanced in the previous year is approximately 50% (0.08 to 0.13), as shown in Figure 7. Notably, the disparity is even more pronounced for an individual who will refinance in one year, as their current quarterly MPC stands at around 0.20. In Appendix B.2, we demonstrate that heterogeneity remains substantial across different shock scenarios among household groups. Refinancers exhibit the smallest immediate consumption increase, which gradually declines over time, whereas no-refinancers experience a larger initial consumption surge, followed by smaller increases in subsequent periods. Overall, when viewed from a policymaker's perspective and considering the available data at the time of fiscal transfers, the analysis provides insights into the potential role for fiscal targeting contingent on the timing of mortgage refinancing, designed to capitalize on varying MPCs.

Figure 7: Quarterly iMPCs Conditional on Home-Equity Extraction Timing



Notes: The iMPCs represent the share of the fiscal transfer that is spent at each period. $T - 1$ and $T + 1$ refer to the timing of refinancing of agents: one year prior and post receiving the fiscal transfer, respectively.

4.5 Savings from Targeted Cash Transfers

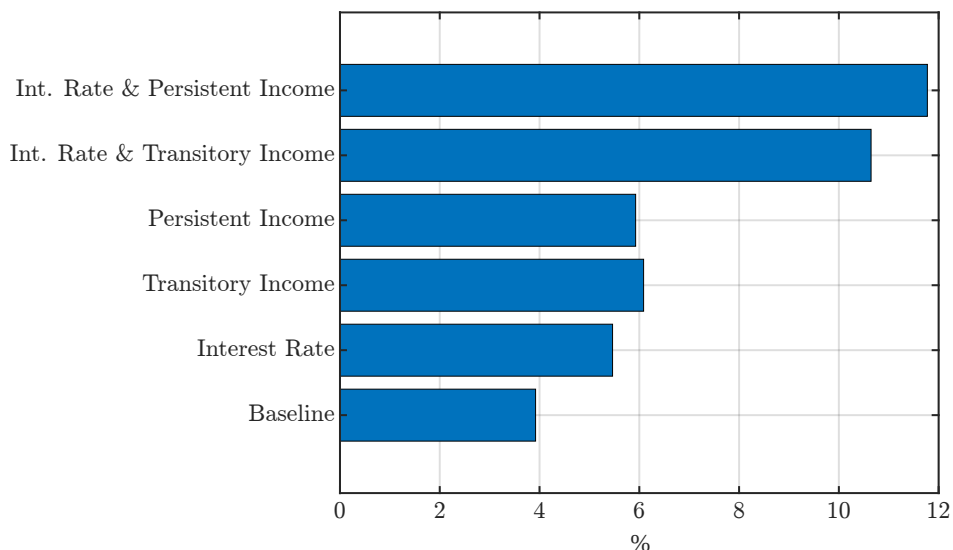
The results from the previous sections suggest a potential role for targeted fiscal programs in terms of efficacy of redistribution towards high MPCs households based on their observable status of no-refinancers in the previous quarters. We examine the magnitudes of potential savings by simulating - for each scenarios analyzed before - the model with cash transfers directed only to households who did not refinance their mortgage within the previous year (i.e. a targeted fiscal program). The total costs of our original program (1) which includes all households in the economy (N) has a total cost of $TC_1 = \tau \cdot N$, where $\tau = \$500$. This program induces a total change in aggregate consumption equal to $\Delta C_1 = \overline{MPC}_1 \cdot \tau \cdot N$, where $\overline{MPC}_1$ is the average MPC in the economy. We then calculate the number of households that are required to generate the same increase in consumption ΔC_1 but considering the average MPC of no-refinancers only ($\overline{MPC}_2$), given by $m' = \Delta C_1 / (\overline{MPC}_2 \cdot \tau)$. Finally, this allows us to compute the total costs for such program (2) as $TC'_2 = m' \cdot \tau$ and related (potential) savings as $S = TC_1 - TC'_2$. Figure 8 shows the savings under all the scenarios as a ratio of the benchmark untargeted program (1): they range from 4% under the baseline economy to approximately 12% in the scenario with interest rate and persistent income drop. Under the assumption that the Covid-19 pandemic can be partially represented by a scenario characterized by an interest rate drop (easing monetary policy) in conjunction with temporary (aggregate) loss in income¹⁰, we leverage on the “Interest Rate & Transitory Income” scenario to impute monetary estimates in potential savings ($\approx 10\%$). These amount to approximately \$30 billion from the \$300 billion passed by the U.S. Congress under the Economic Impact Payments from the CARES (Coronavirus Aid, Relief, and Economic Security) Act.

Our findings align with and provide further insights into previous studies that highlight the relative inefficacy of fiscal transfers during the Covid-19 pandemic (Chetty et al. (2020)). For instance, Kořar et al. (2023) argues that highly leveraged households are more likely to prioritize debt repayment over increased consumption, as they seek to avoid the burden of high future borrowing costs. Similarly, Parker et al. (2022) observes a minimal and statistically insignificant rise in spending during the third round of stimulus payments in March 2021. We view these studies as complementary to our analysis: heavily

¹⁰The loss in income during the Covid-19 pandemic was driven by several factors, such as mass job losses due to business shutdowns, supply chain disruptions and reduction in consumer demand. Here we are not necessarily isolating these channels, but rather taking the scenario as a proxy of severe income shock to all households in the economy.

indebted households not only feel compelled to repay debt rather than consume in the short term, but as they eventually refinance to access liquidity, the effectiveness of fiscal transfers diminishes due to their lower marginal propensity to consume.

Figure 8: Savings from Fiscal Targeting



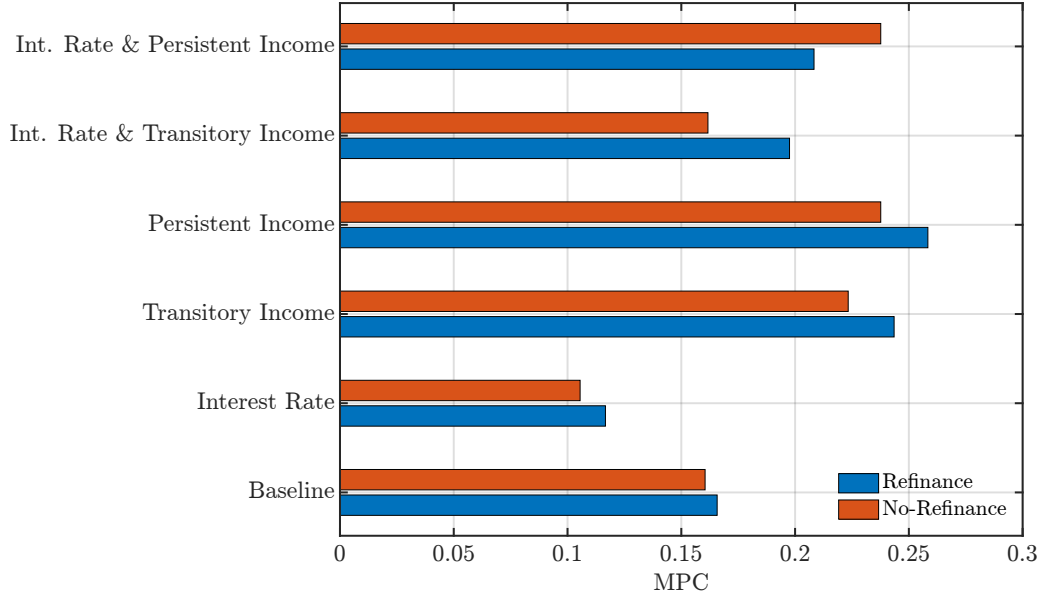
Notes: Savings are expressed in percentages of total costs under an untargeted program.

4.6 Mortgage Refinancing Channel

To assess the impact of mortgage refinancing on fiscal policy, we replicate the simulations under the same scenarios from the previous section within a version of the model in which households lack the ability to access home equity (labelled “No-Refinance”), thereby limiting their avenues for obtaining liquidity to maintain consumption levels when confronted with adverse income shocks. We re-estimate the model by simulated method of moments in order to generate the same targets as the original benchmark in [Boar et al. \(2022\)](#).

Figure 9 shows that MPCs are lower across almost all the simulated scenarios in the alternative re-calibrated version where mortgage refinance is not allowed, with the largest differences occurring when both income and interest rate fall. The only case in which the “No-Refinance” model generates aggregate higher MPC is under the interest rate and persistent income shock scenario, highlighting the potential benefits by accessing home-equity when current and expected income drops. The results are partially driven by the differences in a higher calibrated discount rate that makes households more pa-

Figure 9: Aggregate MPCs - Refinancing Channel



Notes: Quarterly MPCs for each simulated scenarios in the re-calibrated version of the model without mortgage refinancing.

tient. Moreover, the share of hand-to-mouth households in the “No-Refinance” model is lower than the data (37% and 41% at the aggregate, 27% and 32% for homeowners) which strongly impact the overall MPCs in the model. Thus more households have liquid assets available to counteract the negative shocks and to better smooth their consumption, although the distribution of liquid assets is matched fairly close to the data, as in the baseline calibration. The exercise shows the importance of allowing - from a modeling perspective - households to refinance and the extent to which such option relates to aggregate MPCs generated by the model. These dynamics are of paramount importance to study monetary and fiscal policy interactions in heterogeneous agents models which feature liquid and illiquid (e.g. housing) assets, such as in [Kaplan and Violante \(2014\)](#), [Kaplan et al. \(2018\)](#) and [Auclert et al. \(2024\)](#).

5 Conclusions

The composition of households’ balance sheet influences the ability to smooth consumption against adverse shocks, which in turns determines the marginal propensity to consume and thus the efficacy of stabilization policies. In this paper we study the dynamic

effects on MPCs driven by home-equity extraction, which allows households to borrow more against their housing. Based on the methodology by [Blundell et al. \(2008\)](#), we unveil sizeable reduction in the MPCs upon refinancing of around 50%, a result that is robust across different proxies of liquidity constrained households. We leverage on this result to study the importance of mortgage refinancing for fiscal policy efficacy, by showing the presence of a counter-cyclical component both in home-equity extractions as well as in a novel measure of consumption-to-income elasticity that depends on the level of aggregate refinancing activity.

In the last part of the paper, we turn to a partial equilibrium life-cycle model developed by [Boar et al. \(2022\)](#) where households choose between rent and owning a home, with related mortgage choices including refinancing. We use the model to gauge the ability to replicate our micro-estimates of MPCs across the time horizon of refinancing, and to examine the aggregate impact of the heterogeneity among household categories. We show that conditioning on exogenous shocks to income and interest rate, MPCs out of fiscal transfers are in line with empirical estimates from the literature. The heterogeneity in households' MPCs based on their refinancing status suggests a role for fiscal targeting which we examine using simulated data from the model. Our estimates of potential savings in targeted programs range between 4-12% compared to an untargeted program, since no-refinancers have on average higher MPCs. For instance, our estimated savings from the fiscal transfers under the CARES Act of March 2020 amount to approximately \$30 billions.

One of the implication of our work for heterogeneous agents models with housing and assets choices is the importance of the option to “liquidate” part of the home-equity, as this channel is extremely relevant during recession times, where the implications of monetary and fiscal policy are of paramount importance. We show this point in an alternative version of the model where households do not have the option to refinance, in which MPCs are reduced across all the scenarios we consider.

Our findings are based on a partial equilibrium model, which highlights the need for future research to explore the general equilibrium effects of mortgage refinancing. This includes studying both the optimal decision-making processes involved in exercising refinancing options and the subsequent changes in marginal propensities to consume. Further investigation will enhance our understanding of the broader economic impacts of both monetary and fiscal policies.

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Appendix

A Empirical Appendix

This section of the Appendix provides information on the households-level data used for the empirical analysis and additional robustness checks. Section A.1 explains the construction and cleaning of the main households series from the PSID. Section A.2 provides robustness checks for the baseline results. Section A.3 tackles issues related to the estimation strategy of [Blundell et al. \(2008\)](#) and provides additional robustness tests. Sections A.4 and A.5 provide information about the data employed for the counter-cyclicity analysis of Section 3.

A.1 Dataset Construction - PSID

We construct the household-level variables in the PSID database following standard practices. We follow [Kaplan et al. \(2014\)](#) and retrieve information on consumption expenditure as the sum of food at home and food away from home, utilities, gasoline, car maintenance, public transportation, childcare, health expenditures, and education. Household income is the labor earnings of a household plus government transfers, for which we calculate federal and state taxes using the NBER's software TAXSIM to have a measure of after-tax income.

We construct total wealth as the sum of liquid and illiquid assets. Liquid assets are computed as the sum of checking and savings accounts, money market funds, certificates of deposits, savings bonds, and Treasury bills, together with directly held shares of stock in publicly held corporations, mutual funds, or investment trusts. Illiquid assets are the sum of the value of housing, residential and nonresidential real estate net of mortgages and home equity loans, private retirement accounts (such as 401(k)s, IRAs, thrift accounts, and future pensions), cash value of life insurance policies. Net liquid and illiquid assets accounts for liquid debts (e.g. credit cards) and mortgage balances, respectively.

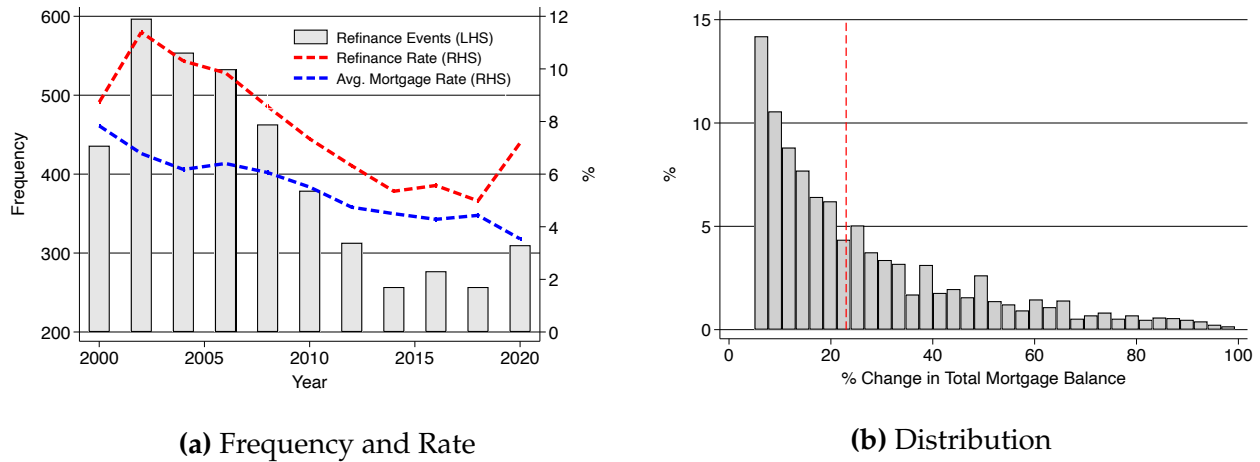
We drop households with missing information on education, state of residence, race, and with income growth higher than 500%, or income drops more than 80%, or below \$100. Top-coded observations on income or consumption are also dropped. Households' head between the age of 25 and 65 are maintained. Our final sample has 68,452 observations over the pooled years 1999-2021 (twelve sample years).

Following the literature, a household has the status of hand-to-mouth (HtM) if the level of net liquid assets is less than two weeks worth of disposable income, assuming a credit limit of one month of income. The HtM status is further split between “poor” and “wealthy” HtM: the former do not hold any illiquid asset, the latter hold positive net amounts. In our sample, 35% of households are “wealthy” HtM, while 25% are “poor” HtM.

We follow the literature (Chen et al. (2020), Boar et al. (2022)) and we assume that a household has refinanced her mortgage if the total balance between two consecutive waves increased by more than 5%. Figure A.1 shows the overall refinancing activity over time and the magnitude of refinancing on the total mortgage balance. In particular, panel (a) is comparable both in magnitude and in terms of dynamics to its counterpart from the Fannie Mae Single-Family Loan-Level (Figure A.3).

Table A.1 provides additional demographic and financial information for a sample split based on households who extracted home-equity in 2001, comparing them to their counterpart of no-refinancers and the average between the two groups. This type of summary statistics is meaningful for a snapshot in our sample given the dynamic effects on households’ balance sheets.

Figure A.1: Refinancing Activity - PSID



Notes: panel (a) plots the frequency (lef-hand-side axis) of refinancing events for each year, the annualized rate of aggregate refinancing over the total mortgages and the average mortgage rate (right-hand-side axis) for each survey wave. Panel (b) plots the distribution of the percentage changes in total mortgage balance, conditioning on refinancing. The dashed red line indicates the median value change.

Table A.1: Summary Statistics

	Refinancers		Non-Refinancers		Sample	
Age	45.5	(8.3)	44.3	(9.6)	44.4	(9.4)
College Education (%)	60.0	(50.0)	60.0	(50.0)	60.0	(50.0)
Employed (%)	92.0	(27.6)	90.0	(30.0)	90.2	(29.7)
Unemployed (%)	2.3	(15.1)	2.0	(14.1)	2.0	(14.2)
Retired (%)	3.6	(18.8)	5.3	(22.4)	5.0	(21.9)
Family Size	3.3	(1.3)	3.1	(1.4)	3.1	(1.4)
Number of Children	1.0	(1.1)	1.0	(1.2)	1.0	(1.2)
Support Children Outside Family (%)	12.6	(33.2)	14.8	(35.5)	14.5	(35.2)
Disposable Income (\$)	77,536	(35,124)	70,888	(35,029)	71,869	(35,112)
Extra Income (%)	38.1	(48.6)	29.4	(45.6)	30.7	(46.1)
Net Liquid Assets (\$)	18,400	(48,169)	23,105	(56,112)	22,409	(55,022)
Net Illiquid Assets (\$)	98,713	(107,139)	119,180	(136,002)	116,153	(132,302)
Net Liquid Assets over Income	0.17	(0.51)	0.32	(1.03)	0.30	(0.97)
Debt-to-Income Ratio	1.53	(1.06)	1.03	(0.97)	1.10	(1.0)
Other Assets (\$)	2,004	(6,104)	1,790	(5,761)	1,822	(5,812)
<i>N</i>	302		1737		2039	

Notes: Values are means, with standard deviations in parentheses. Reference year is 2001. Data are deflated by the CPI deflator (2006 base year) where applicable.

A.2 Refinancing MPCs - Robustness checks

A.2.1 Liquidity Constrained

In this section we perform several robustness checks for our main result on MPCs conditional on a refinancing event. We consider several instances that might have incentivized a household to refinance in order to access liquidity. These include cases in which the change of medical or educational expenses were positive in conjunction with the refinancing event (i.e. if the household refinances the mortgage at time t , the change in expenditure between $t - 1$ and t is positive). Furthermore, we consider proxies for liquidity

constraints as the “wealthy” HtM indicator, as well as if households have liquid wealth-income ratio below the median value and debt-service-income ratio above the median value (Cooper (2013)).

Compared to the benchmark results, here we leverage on the average MPCs before and after refinancing, as we focus on smaller portions of our sample, and we report the results in Table A.2. The changes are significant and comparable to our main result, ranging from 18% to 26%. The only instance in which the drop in MPCs is not statistically significant is when we consider households with large debt-service-income ratio, as these potentially relates to pre-committed expenditures in the short-medium run that do not correlates with the liquidity need behind the refinancing choice. The smallest number of observations in that scenario corroborates our intuition.

Table A.2: Change in MPCs upon Refinancing

	Baseline	W-HtM	Medical	Educational	Liq. Wealth/Income	Debt-Service
ΔMPC	-0.20** (2.04)	-0.26** (2.16)	-0.24** (2.17)	-0.25** (2.31)	-0.18* (1.72)	-0.19 (0.94)
N	5,094	3,009	3,658	4,184	4,009	2,044

Notes: t-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.2 Interest Rate Incentives

The recent literature on the mortgage refinancing channel of monetary policy has stressed the role of interest rate incentives as the key mechanism through which changes in policy rate can affect consumption and the economy (Wong (2021), Berger et al. (2021), Eichenbaum et al. (2022)). We confirm that our main results are not affected by benefits driven by changes in interest rates that households could obtain when refinancing. Table A.3 shows the average MPC before and after refinancing for a subset of households that do not report a change in their mortgage rate. The change in MPC, in this scenario, is fairly large (0.32) and significant, which strengthen the liquidity channel as the main mechanism behind our results.

Table A.3: (Bi-annual) MPCs Estimates over Refinancers - No Interest Rate Incentives

	Before Refinance (1)	After Refinance (2)
MPC	0.47*** (0.11)	0.15* (0.08)
<i>N</i>	3,439	3,439

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. H_0 : MPC (1) = MPC (2), p-value: 0.02

A.2.3 Unemployment

A major source of income volatility which affects consumption growth is the risk of unemployment. Here we test that our results are not driven by refinancers who lost their job, both at the same time (t) or two years ($t - 1$) before refinancing.¹¹ Indeed, the sample of unemployed households who refinance is extremely small, and the MPCs estimates are not statistically significant. Furthermore, constraints such as the PTI reduce the likelihood that individuals without employment are able to write new mortgage contracts, reinforcing the idea that households who opt to cash-out home equity do so mainly because of liquidity needs and not due to large and severe income shocks.

Table A.4: (Bi-annual) MPCs Estimates over Refinancers - Unemployment

	Before Refinance (1)	After Refinance (2)
MPC	0.16 (0.26)	0.00 (0.13)
<i>N</i>	724	724

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

¹¹Here we control for contemporaneous refinancing event and unemployment status, but the results are invariant when we consider unemployment in the preceding two years of refinancing. Results are available upon request.

A.2.4 Male Household Head

We further control for low workforce engagement by restricting our sample to households with male heads. Table A.5 shows that our results remains robust to this specification.

Table A.5: (Bi-annual) MPCs Estimates over Refinancers - Male Households' Heads

	Before Refinance (1)	After Refinance (2)
MPC	0.44*** (0.08)	0.19*** (0.06)
<i>N</i>	4,641	4,641

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. $H_0 : \text{MPC (1)} = \text{MPC (2)}$, p-value: 0.02.

A.2.5 Cash-on-hand

Table A.6 reports the MPCs estimates over cash-on-hand, a widely used measure of liquidity which adds monthly labor income to the available liquid assets (Jappelli and Pistaferri (2020)). The change in MPCs upon refinancing for households below the median of the cash-on-hand distribution confirms sizeable drop in MPCs for potentially liquidity constrained households, similarly to Table A.2. At the same time, the average MPCs of refinancers below the median value of the cash-on-hand in the sample remains higher than the one in the top of the distribution, in line with Table 3.

Table A.6: MPC over Refinancers Households - Cash-on-hand

	Change Cash-on-hand < Median (1)	Average Cash-on-hand < Median (2)	Average Cash-on-hand > Median (3)
$\Delta \text{MPC}/\text{MPC}$	-0.21*** (0.06)	0.23*** (0.04)	0.18*** (0.06)
<i>N</i>	3,692	6,515	2,489

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. $H_0 : \text{MPC (2)} > \text{MPC (3)}$, p-value: 0.77.

A.2.6 Total Consumption

The baseline results are based upon changes in non-durable consumption, thus it is interesting to check whether our results hold when we include durable goods (vehicles and household furnishings and equipments) in the measure of consumption. The change in MPCs is robust when we consider total consumption, while durable goods expenditure does not generate qualitatively significant differences. Thus non-durable goods expenditure drive the key baseline result, reinforcing the idea that refinancing allows to relax liquidity constraints for immediate consumption needs, lowering therefore MPCs.

Table A.7: Change in MPCs upon Refinancing - Total Consumption and Durable Goods

	Total Consumption	Durable Goods
ΔMPC	-0.21** (1.95)	-0.25 (0.90)
N	5,131	4,915

Notes: t-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Blundell et al. (2008) - Robustness checks

In this section we perform several robustness checks that relate to the methodology by [Blundell et al. \(2008\)](#), in particular the exogeneity assumption of no advance information which could be violated in the setting of mortgage refinancing.

A.3.1 Variance-Covariance Matrix of Income and Consumption Growth

We start by estimating the variance-covariance matrix of current income and consumption growth with future income growth, conditioning on refinancing-relevant periods, as in our main specification ($t \in [\text{Refi} - 2, \text{Refi} + 2]$). Table [A.8](#) informs the length of the possible anticipation effect. In short, the transitory shocks are short-lived, as average auto-covariance of detrended income growth is statistically significant up to time $t + 1$. This implies the transitory shocks are represented by an $MA(0)$ process. The covariance between detrended consumption growth is significant up to time $t + 1$ as well, which departs from a random walk. This exercise implies that we can estimate a robust version of the BPP estimator that relaxes the random walk assumption on log consumption, as in [Commault \(2022\)](#). When the transitory shocks are short-lived, a single value of future

detrended income growth can serve as an instrument that matches the robust estimator's requirements. We currently adopt the same instrument in the second stage of our BPP baseline estimation.

Table A.8: Variance-Covariance Matrix of Income And Consumption Growth

	$\text{cov}(\Delta\hat{\varepsilon}_{i,t}^y, .)$	p-value	$\text{cov}(\Delta\hat{\varepsilon}_{i,t}^c, .)$	p-value
$\Delta\hat{\varepsilon}_{i,t}^y$	0.102	0.000	0.113	0.000
$\Delta\hat{\varepsilon}_{i,t+1}^y$	-0.037	0.000	-0.013	0.000
$\Delta\hat{\varepsilon}_{i,t+2}^y$	-0.002	0.153	0.001	0.277
$\Delta\hat{\varepsilon}_{i,t+3}^y$	-0.003	0.035	-0.001	0.892

Notes: First differenced residuals of both log consumption and log income were estimated based on equation (7). For the first entry of the first column, we test for if the variance of $\Delta\hat{\varepsilon}_{i,t}^y$ is larger than 0.0001 for completeness. Other p-values reflect two-tailed t-tests where we test the hypothesis $\text{cov}(\Delta\hat{\varepsilon}_{i,t}^y, \Delta\hat{\varepsilon}_{i,t+h}^{y/c})/\text{var}(\Delta\hat{\varepsilon}_{i,t}^y)$ is different from 0. We perform a two-tailed t-test here due to the fact that we cannot assume the sign of the covariance. If consumption or income growth follows a random walk, the covariances could be positive or negative, hence we perform two-tailed t-tests.

A.3.2 Measurement Errors

The BPP methodology faces several biases in its estimation process, largely due to the biannual structure of the dataset. These biases arise from factors such as the correlation between past shocks and individual-specific borrowing conditions, which negatively impacts the relationship between income and consumption growth. Additionally, the timing of shocks influences consumption growth initially, but this effect diminishes over time, leading to downward-biased estimates. Again this background, we assume that the measurement error variance is large as the variance of transitory income shocks within a single wave of the survey.¹² Table A.9 confirms the baseline result.

¹²In practice, we re-estimate MPCs using the BPP method using $\Delta\hat{\varepsilon}_{i,t}^y/2$ in the first stage.

Table A.9: (Bi-annual) MPCs Estimates over Refinancers - Measurement Errors

	Before Refinance (1)	After Refinance (2)
MPC	0.80*** (0.16)	0.41*** (0.10)
N	5,094	5,094

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. H_0 : MPC (1) = MPC (2), p-value: 0.05

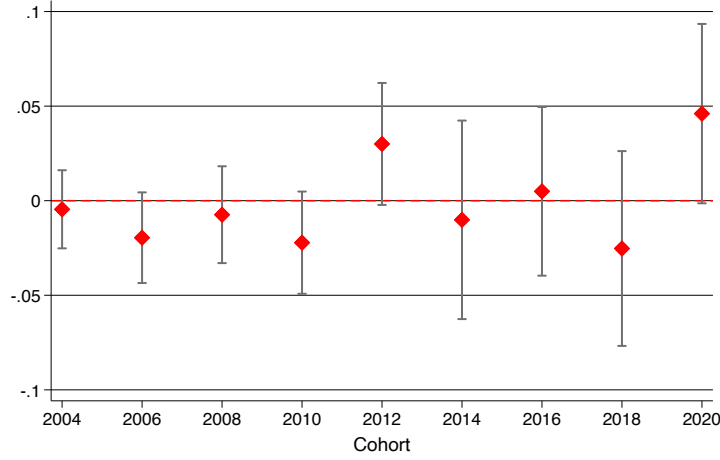
A.3.3 Anticipations

One of the assumptions of the BPP method is that households do not have advanced information about future shocks. The violation of this assumption concerns - in our context - the likelihood of anticipating the refinancing event two years ahead of time. First, based on unconditional data, we see that the correlation between the de-trended series of income shocks ($\Delta y_{i,t}$) and the our lagged refinancing indicator ($\text{Ref}_{i,t-1}$) is positive but extremely small (0.01), which is reassuring of the fact that households do not receive systematically large shocks in the two years preceding the cash-out of home equity. Second, we test for anticipatory effects on log consumption by cohorts, explicitly into account the underlying assumptions behind the staggered event-study of Section 2.4. To this extent we run the following specification:

$$\log c_{i,t} = \alpha_i + \alpha_t + \sum_{s=1}^T \sum_{r=1}^R \beta_{s,r} T_{i,s} \cdot C_{r,t} + \mathbf{X}_{i,t}' \gamma + \varepsilon_{i,t}$$

where α_i and α_t are time and households fixed effects, $\mathbf{X}_{i,t}$ is the standard set of demographics controls, and $T_{i,s} \cdot C_{r,t}$ is an interaction term between two dummy variables: $T_{i,s}$ refers to a year s in our panel for household i , and $C_{r,t}$ refers to a refinancing cohort r at time t . Intuitively, $\beta_{s,r}$ estimates both anticipatory and post treatment effects on consumption growth by cohort of refinancers. Figure A.2 plots the average estimate for each cohort r for all the time t prior to the refinancing event. The only cohorts for which there is a statistically significant anticipatory effect are the ones who refinanced in 2012 and in 2020. We then re-estimate our baseline scenario excluding these cohorts, and confirm in Table A.10 that our results are not driven by this subset of households.

Figure A.2: Average Anticipatory Effect by Refinancing Cohort



Notes: average anticipatory effect estimates by cohort in red diamonds, while grey bars indicate 95% confidence intervals. Each estimate is the average of all $\beta_{s,r}$ for a given cohort r and for all $s < r$.

Table A.10: (Bi-annual) MPCs Estimates over Refinancers - Exclude 2012/2020 Cohorts

	Before Refinance (1)	After Refinance (2)
MPC	0.44*** (0.11)	0.19*** (0.06)
N	3,657	3,657

Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. H_0 : MPC (1) = MPC (2), p-value: 0.06.

A.3.4 AR(1) Persistent Component

The BPP approach assumes that income shocks are either permanent or transitory. We test our baseline result in the case in which shocks are persistent, following an AR(1) process:

$$p_{i,t} = \rho \cdot p_{i,t-1} + \eta_{i,t}$$

We estimate MPCs over refinancing assuming that the difference in income is now given by $\Delta \tilde{y}_{i,t} = y_{i,t} - \rho \cdot y_{i,t-1}$, following [Kaplan and Violante \(2010\)](#). Table confirms our baseline result under several cases we consider for the value of the persistent parameter.

Table A.11: MPC over Refinancers Households - AR(1) Persistent Component

	$\rho = 0.80$	$\rho = 0.90$	$\rho = 0.95$
ΔMPC	-0.31** (0.15)	-0.23** (0.04)	-0.21** (0.11)
N	5,094	5,094	5,094

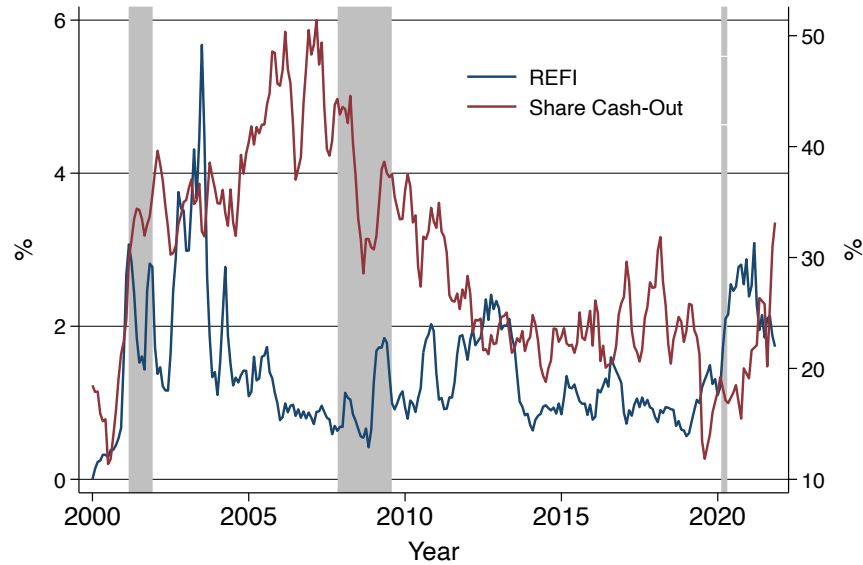
Notes: Bootstrapped standard errors based on 250 replications in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Fannie Mae Data

We use loan-level data for the U.S. from the Fannie Mae Single-Family Loan-Level historical dataset.¹³ Fannie Mae is one of the largest agency mortgage-backed securities (MBS) that are government-sponsored enterprises (GSE), together with Freddie Mac, and Ginnie Mae. By the end of 2022, agency MBS accounted for 67.8% (\$8.9 trillion) of the total outstanding mortgage debt, while by March 2023, outstanding securities in the agency market totaled \$8.8 trillion, 41.1% (\$3.6 trillion) of which was Fannie Mae (Goodman et al. (2023)). The panel contains information about several loan-specific characteristics at origination for fixed-rate mortgage, however we do not rely on these additional information, as these would aim to capture degree of heterogeneity in refinancers which is not the focus of our paper. Each loan in the dataset is uniquely identified at origination with an identification number, which does not allow to track over time the same borrower (household) with multiple mortgage contracts, thus we treat each loan as owned by a new borrower. The sample spans from January 2000 to November 2021, we keep only fixed-rate contracts and for each quarter we select a random set of 10% of newly originated loans (in order to limit the computational burden). We define refinanced those mortgages that are prepaid voluntarily before maturity and focus on total refinancing, and we construct an aggregate index as the share of prepaid mortgages over the total number of active mortgage in our sample in a given month. Figure A.3 plots the REFI_t index, together with the share of cash-out loans that we use as robustness check (see Section A.6): refinancing activity was extremely large in the years preceding the Great Financial Crisis driven by the housing boom, and it experienced a large spike in the wake of the Covid-19 pandemic (co-founding liquidity needs and historically low mortgages rates).

¹³Data available at [Fannie Mae Single-Family Loan-Level website](#).

Figure A.3: Fannie Mae Refinance Indices

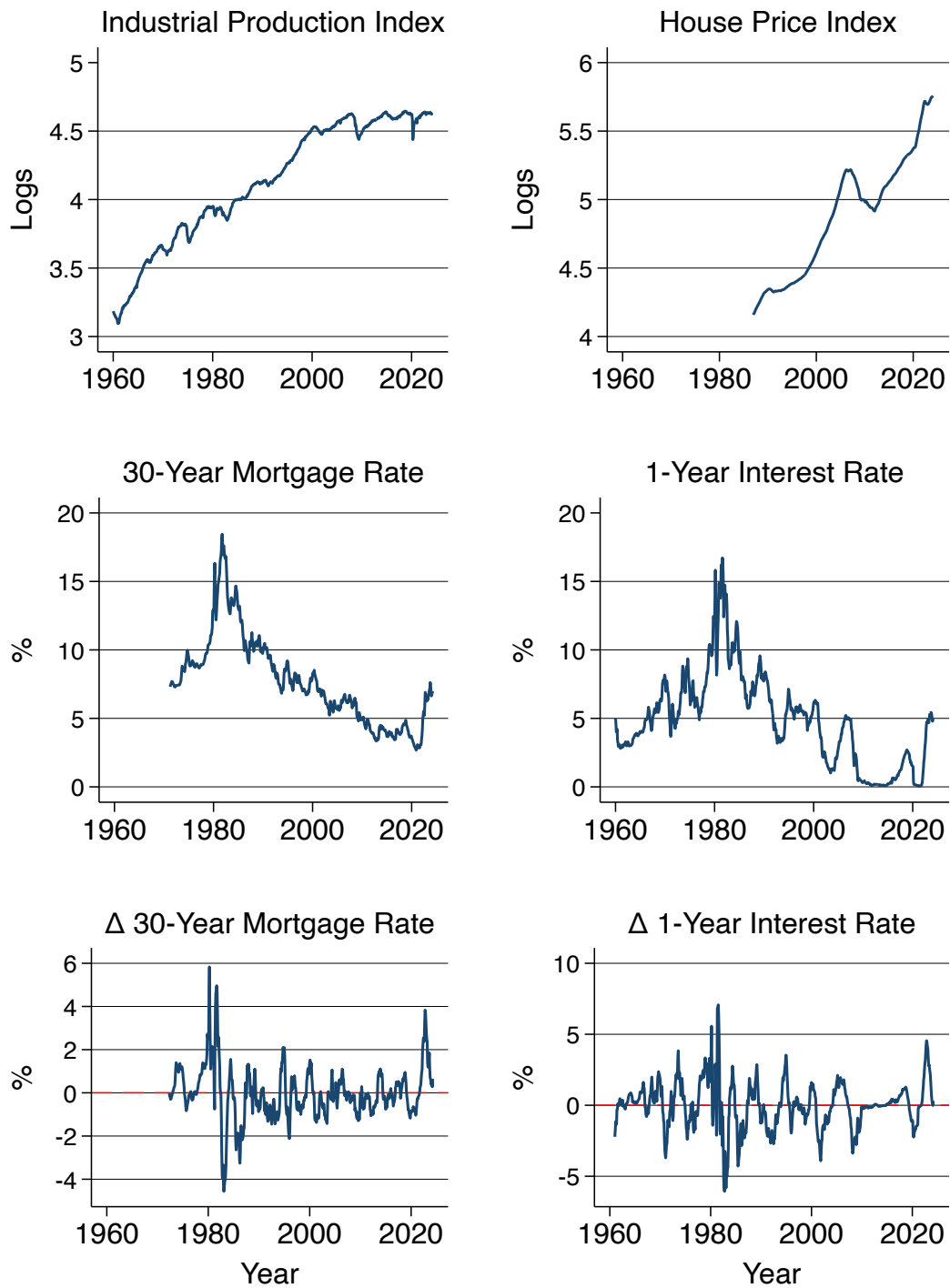


Notes: The gray shaded areas show the U.S. recessions according to the NBER. See main text for the construction of the refinance index and the share of cash-out loans.

A.5 Macroeconomic Data

We retrieve macroeconomic time series from the Federal Reserve Economic Data (FRED) Database managed by the Federal Reserve Bank of St. Louis. We use the Industrial Production: Total Index (**INDPRO**), the S&P CoreLogic Case-Shiller U.S. National Home Price Index (**CSUSHPIA**), the 30-Year Fixed Rate Mortgage Average in the U.S. (**MORTGAGE30US**), and the Market Yield on U.S. Treasury Securities at 1-Year Constant Maturity (**GS1**). We apply the logarithm function to indices, while rates (in level and change) are in percentages.

Figure A.4: Macroeconomic Data



A.6 Counter-cyclical Mortgage Refinancing

We confirm that the counter-cyclical feature of mortgage refinancing is robust to a sample split based on the Great Financial Crisis (Table A.12), representing a structural break in the housing market and on the interest rates path.

Table A.12: Aggregate Refinancing and Economic Activity - GFC Sample Split

	Before GFC			After GFC		
Dependent variable: $\log(\text{REFI}_t)$	(1)	(2)	(3)	(1)	(2)	(3)
$\log(\text{IP}_t)$	-7.25*** (0.96)	-18.51*** (1.50)	-12.36*** (1.86)	-2.77*** (0.75)	-1.83** (0.77)	-3.62*** (0.60)
$\log(\text{HPI}_t)$		3.77*** (0.30)	2.85** (0.78)		0.47*** (0.17)	-0.63*** (0.17)
ΔRM30_t		-0.54*** (0.06)	-0.51*** (0.14)		-0.34*** (0.05)	-0.01 (0.05)
Controls	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$
N	94	94	94	148	148	148
R^2	0.19	0.81	0.83	0.09	0.33	0.74

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To test further our assumption of prepaid mortgages in Fannie Mae as a good proxy of mortgage refinancing, we employ the information provided in the dataset about the purpose of the newly originated loans each quarter. We calculate the share of cash-out mortgages as the ratio of loans labeled as "Cash-Out Refinance" to the total number of new loans entering the dataset in a given quarter. This measure differs from the baseline, which only considers refinanced mortgages among the most recent loans. In contrast, the REFI_t indicator measures the ratio of prepaid loans to the total existing loans within a given quarter. Table A.13 reports the estimates in this alternative scenario, which confirms the counter-cyclical of cash-out refinance in the specifications without controls.

Table A.13: Aggregate Share of Cash-Out and Economic Activity

Dependent variable: $\log(\text{Cash-Out Share}_t)$	(1)	(2)	(3)
$\log(\text{IP}_t)$	-1.31*** (0.34)	-2.10*** (0.69)	0.48 (0.45)
$\log(\text{HPI}_t)$		0.24 (0.18)	1.48** (0.15)
ΔRM30_t		0.03 (0.04)	-0.26*** (0.03)
Controls	X	X	✓
N	254	254	254
R^2	0.05	0.06	0.50

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.7 Consumption-to-Income Elasticity

Table A.14 reports the estimates of the regression 12 based on PSID data. In a second specification we include as controls the aggregate refinancing level (Ref_t) and interest rate gap ($\Delta i_t = i_t^{\text{PSID}} - i_t^{\text{Market}}$), measured as the difference between the average mortgage rates in the PSID and the current market average (nation wide) mortgage rate in the previous period. Thus, the consumption-to-income elasticity $\varepsilon_{i,t}^c$ is computed as:

$$\varepsilon_{i,t}^c \equiv d\log c_{i,t} / d\log y_{i,t} = \beta + \gamma \cdot \text{Ref}_{i,t} = 0.32 - 0.03 \cdot \text{Ref}_{i,t}$$

By averaging over time i the refinancing indicator in a given survey wave t (i.e. a measure of refinancing share, as in Figure A.1), we calculate the consumption-to-income elasticity ε_t^c as function of aggregate home-equity extraction Ref_t .

Table A.14: Consumption-Income Sensitivity to Refinancing Activity

Dependent variable: $\log c_{i,t}$	(1)	(2)
$\log y_{i,t}$	0.32*** (0.00)	0.33*** (0.01)
$\text{Ref}_{i,t}$	0.04*** (0.00)	0.03*** (0.00)
$\log y_{i,t} \cdot \text{Ref}_{i,t}$	-0.03* (0.02)	-0.04** (0.02)
Controls	X	✓
N	35,687	14,688
R^2	0.23	0.22

Notes: standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Model Appendix

This section of the Appendix reports additional information about the simulations from the model used in the main text. For further details on the model itself, see [Boar et al. \(2022\)](#).

B.1 Simulations

Each scenario described in Section 4.3 is based on a baseline simulated economy populated by 25,000 households, for 244 quarters (61 years of life-cycle). Against this baseline economy (labelled M^b), we simulate an exogenous shock to a parameter x and generate an alternative simulated version (labelled M^x). When we calculate MPCs in the model we retrieve the simulated economy and apply an exogenous (fully unanticipated) transfer that enters the budget constraint of the agents in a given time period. We compute the MPCs for each scenarios as follows:

$$MPC_{i,t}^x = \frac{\tilde{C}_{i,t}^x - C_{i,t}^x}{\tau}$$

where $\tilde{C}_{i,t}^x$ is the level of consumption when households receive the fiscal transfer. We average over households i by groups and over the entire population to obtain MPCs

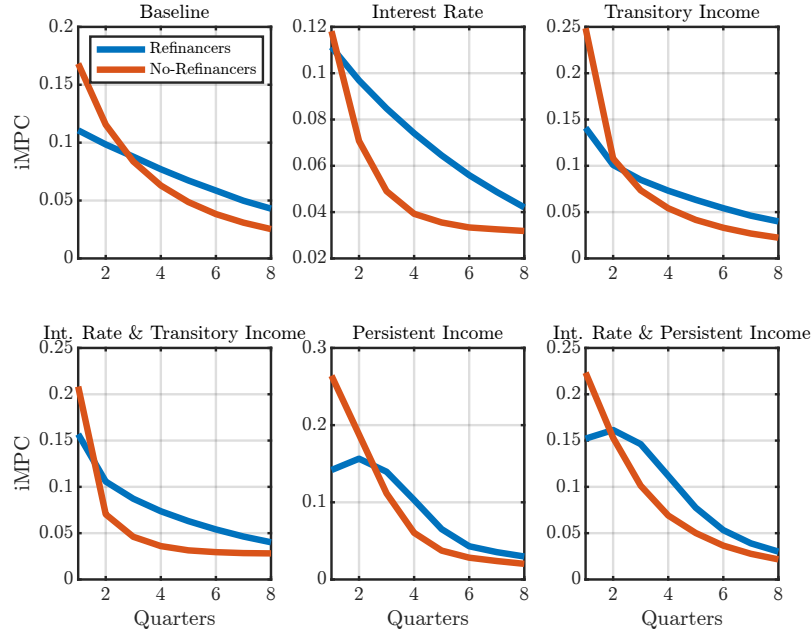
by refinancers, no-refinancers and at the aggregate level. In the cases in which two parameters are shocked, we generate a simulated economy with the first shock (M^{x_1}) and later simulate a second shock on top of it (M^{x_1, x_2}). We identify agents as no-refinancers in the shocked scenarios as targets for our fiscal experiments and compare them to an un-targeted fiscal program (i.e. agents who are not refinancing their mortgage would continue to keep their status of no-refinancers upon receiving the transfer).

B.2 Intertemporal MPCs

In our analysis, fiscal transfer simulations enable us to examine how different household groups adjust their consumption in response to an exogenous increase in disposable income. We focus exclusively on fully unanticipated transfers, in contrast to income changes expected to occur in future periods. In our iMPCs setup, we consider only the first column, $M_{t,0}$. Figure B.1 illustrates the average iMPCs for refinancers versus non-refinancers under each simulated shock: households who refinanced within the year prior to the transfer show, on average, a lower propensity to consume compared to non-refinancers. The latter group experiences a larger immediate consumption increase, reflecting minimal consumption smoothing given their shorter planning horizon and greater immediate benefit from consumption.

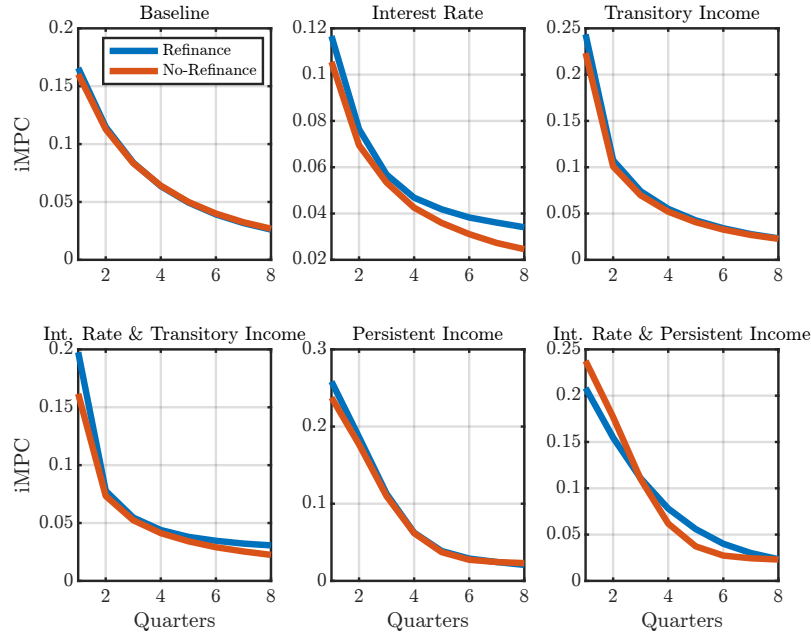
Figure B.2 compares the iMPCs at the aggregate level between the baseline model and the alternative without the option to refinance mortgages. The observed differences are minimal, primarily affecting the immediate response (as detailed in Section 4.6 of the main text). These findings suggest that, while refinancing has a limited aggregate impact on fiscal policy efficacy, it provides certain households with a tool to smooth consumption in response to adverse shocks.

Figure B.1: Quarterly iMPCs - Simulated Scenarios - Refinancers and No-Refinancers



Notes: Quarterly iMPCs for each simulated scenarios across households groups. The iMPCs represent the share of the fiscal transfer that is spent at each period.

Figure B.2: Quarterly iMPCs - Simulated Scenarios - Refinancing Channel

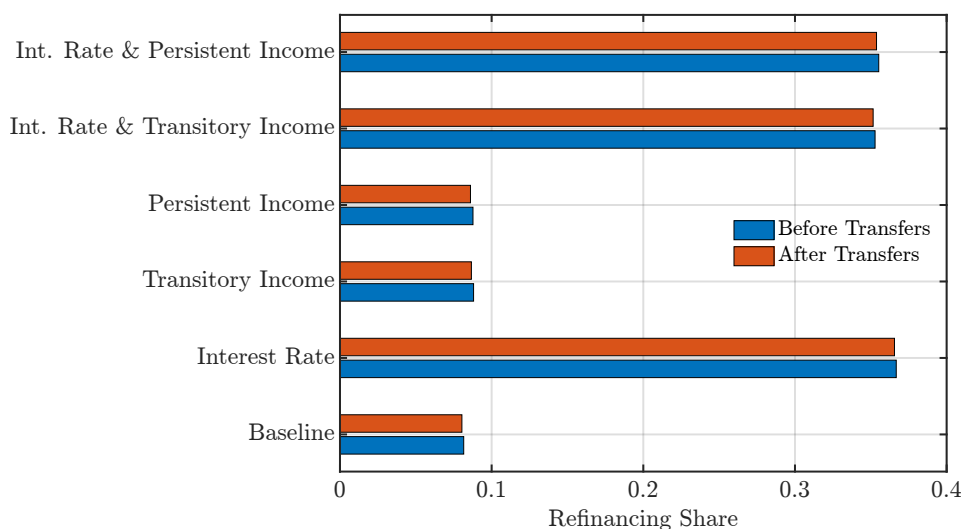


Notes: Quarterly iMPCs for each simulated scenarios in the re-calibrated version of the model without mortgage refinancing. The iMPCs represent the share of the fiscal transfer that is spent at each period.

B.3 Share of Refinancers

For each simulated shock scenario, we calculate the overall share of refinancers in the economy and compare it before households receive fiscal transfers with the ex-post simulated economy including these transfers. This approach allows us to identify the primary drivers of refinancing activity and examine whether households can marginally access home equity upon receiving additional—albeit modest—funds. Figure B.3 shows that mortgage rates are the principal factor influencing refinancing, with rates increasing from an average of 8% in the baseline scenario to around 35% for cases involving a drop in interest rates. Moreover, fiscal policy has minimal impact on aggregate refinancing activity, as differences between scenarios with and without transfers are negligible. These findings support the idea that during recessions or economic downturns, when monetary easing prompts some households to extract home equity, targeted fiscal programs can enhance policy efficacy, as discussed in the main text.

Figure B.3: Refinancing Shares - Simulated Scenarios



Notes: The refinancing shares are calculated as the ratio of the number of refinancers to the total number of homeowners with an active mortgage at a given time t . This same period t serves as the reference point for when fiscal transfers are distributed.